Geochimica et Cosmochimica Acta 419 (2026) 225–240

Contents lists available at ScienceDirect

Geochimica et Cosmochimica Acta

journal homepage: www.elsevier.com/locate/gca

Flow-driven regulation of $CaCO_3$ precipitation and fracture evolution in heterogeneous fractures

Tengfei Ma [a], Hang Deng [b,c], Puxuan Kang [a], Yuchao Feng [a], Xiaofan Yang [d], Liang Zhao [e], Xueying Li [d], Dexi Wang [f], Xingming Chen [f], Zhengmao Yang [f], Weisheng Qiu [f], Junfeng Ji [a], Yuanyuan Liu [g,*]

[a] MOE Key Laboratory of Surficial Geochemistry, School of Earth Sciences and Engineering, Nanjing University, Nanjing 210023, China
[b] School of Mechanics and Engineering Science, Peking University, Beijing, China
[c] Department of Energy and Resources Engineering, Peking University, Beijing, China
[d] State Key Laboratory of Earth Surface Processes and Resource Ecology, Faculty of Geographical Science, Beijing Normal University, Beijing, China
[e] Frontier Science Center for Critical Earth Material Cycle, School of Sustainable Energy and Resources, Nanjing University, Nanjing 210023, China
[f] Sinopec East China Oil & Gas Company, Nanjing, China
[g] State Key Laboratory of Critical Earth Material Cycling and Mineral Deposits, School of Earth Sciences and Engineering, Nanjing University, Nanjing 210023, China

ARTICLE INFO

Associate editor: Carl Steefel

Keywords:
CO_2 sequestration
Fracture evolution
Carbonate precipitation
Random nucleation
Polymorphs

ABSTRACT

$CaCO_3$ precipitation plays a pivotal role in fracture evolution and permeability in subsurface reservoirs, directly impacting the efficiency and reliability of geological carbon sequestration. Yet, how flow-driven transport interacts with nucleation, polymorph deposition, mineral transitions, and fracture heterogeneity remains poorly constrained. Here, we combine micromodel experiments, in situ Raman spectroscopy, pore-scale fluid and solute transport simulations, and Darcy-scale advection–diffusion–nucleation simulations to unravel coupled dynamics of flow, mass transfer, and mineral precipitation. We show that flow-controlled mixing plays a critical role on early-stage nucleation, polymorph precipitation, and subsequent fracture structure evolution. In most regions, supersaturation develops progressively through solute mixing, triggering initial calcite nucleates, followed by the accumulation of metastable $CaCO_3$ phases that fill pore space and enhance the total precipitation mass. However, at high flow rates, reactants advected from finer fractures into coarser, high-velocity preferential flow paths form confined supersaturation zones along streamlines. These zones promote rapid amorphous calcium carbonate (ACC) formation and delay its transformation to stable calcite. These early-stage processes—including stochastic nucleation, transient ACC persistence, and polymorph precipitation and transitions—do not alter the final mineralogy but strongly regulate fracture geometry, saturation fields, and the reproducibility of precipitation. As a result, divergent long-term outcomes emerge: high flow rates restrict precipitation to narrower zones, producing reproducible but clog-prone fracture sealing; whereas low flow rates broaden saturation zones, amplify the stochastic nature of nucleation, and increase variability in precipitation distribution and fracture geometry evolution. Our findings reveal an underexplored mechanism by which flow-controlled mixing governs early-stage nucleation and polymorph precipitation, thereby regulating long-term fracture structure evolution and precipitation variability. These insights provide a mechanistic framework for optimizing CO_2 injection strategies, predicting reservoir heterogeneity, and improving models of mineral trapping in fractured subsurface systems.

1. Introduction

Mineral precipitation and dissolution are fundamental processes governing subsurface geological evolution, with significant implications for natural systems and engineered operations such as geological carbon sequestration (GCS) and geothermal energy production (Bashir et al., 2024; De Silva et al., 2015; Gislason and Oelkers, 2014; Lu et al., 2024; Zeng et al., 2023). In particular, carbonate precipitation is a key geochemical process in GCS, influencing both CO_2 mineralization efficiency and the evolution of reservoir properties (Matter et al., 2016; White et al., 2020). The precipitation of carbonate contributes to long-term CO_2 storage by converting dissolved CO_2 into stable carbonate

* Corresponding author.
E-mail address: yuanyuan.liu@nju.edu.cn (Y. Liu).

https://doi.org/10.1016/j.gca.2026.03.001
Received 30 September 2025; Accepted 1 March 2026
Available online 5 March 2026
0016-7037/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

minerals, but it also modifies the porosity and permeability of the storage formation, which in turn affects the overall extent of GCS (Olson et al., 2009; White et al., 2020; Xu et al., 2017). Fractures, which are ubiquitous in subsurface environments and frequently created or enhanced through fracturing techniques, play a crucial role in controlling CO_2 transport and mineralization dynamics (Laubach et al., 2019; Nisbet et al., 2024; Raza et al., 2022; Viswanathan et al., 2022). With growing demand for large-scale carbon storage, fracturing techniques have become essential for enhancing formations permeability and facilitating CO_2 injection and storage (Gupta and Verma, 2023; Lyu et al., 2021; Nisbet et al., 2024; Soulaine et al., 2021). However, while significant progress has been made in studying mineral precipitation and dissolution in pore networks, research in fractured systems remains comparatively limited (Steefel and Hu, 2022). The complex interplay between flow, mass transfer, and precipitation kinetics in fractures introduces unique challenges, as fractures exhibit heterogeneous geometries (Anders et al., 2014), dynamic flow regimes (Neuman, 2005; Shapiro et al., 2007), and localized chemical conditions (Shapiro et al., 2007; Waber and Smellie, 2008) that significantly differ from pore networks. This research gap is particularly relevant for applications of CO_2 mineral trapping efficiency, fracture evolution, and long-term reservoir integrity.

Natural fractures typically exhibit complex heterogeneity, characterized by an interwoven network of high- and low-permeability fractures (Sanglas et al., 2024). This structural variability gives rise to distinct mixing regimes, with advection-dominated transport along preferential flow paths leading to early breakthrough, and diffusion-dominated transport into low-permeability zones causing prolonged tailing (Sanglas et al., 2024; Shapiro, 2001; Geiger et al., 2010). To evaluate the relative importance of advection, diffusion, and reaction in such systems, dimensionless numbers such as the Reynolds (Re), Péclet (Pe), and Damköhler (Da) numbers are commonly employed (Ahoulou et al., 2020; Wang and Yu, 2018; Wetzel et al., 2022; Zheng et al., 2025). The applicability of such dimensionless frameworks, however, is strongly influenced by system complexity and spatial scale (Li et al., 2008; Soulaine et al., 2017; Starchenko and Ladd, 2018). In idealized systems—such as uniform or isolated irregular fractures—where flow fields are well-defined, dimensionless numbers provide valuable mechanistic insights (Deng et al., 2018; Soulaine et al., 2017; Yang et al., 2024). Accordingly, many studies have focused on such simplified fracture frameworks (Ahkami et al., 2020; Noiriel et al., 2013; Song et al., 2014; Xiao et al., 2025). Yet these systems fail to capture the heterogeneity, anisotropy, and dynamic mixing conditions characteristic of natural fracture media. To address this gap, a recent study employed synthetic heterogeneous micromodels to extend such investigations to more realistic geometries, specifically probing $CaCO_3$ precipitation across varying Pe–Da regimes (Nguyen et al., 2025). Despite this advancement, efforts to generalize mineral phase and morphology based solely on Pe and Da have led to divergent interpretations. For instance, Tan et al. (2016) report that high Da favors dendritic $CaCO_3$ formation, whereas Nguyen et al. (2025) observed dendritic morphologies under low Da (Nguyen et al., 2025; Tan et al., 2016). These discrepancies underscore the limitations of using dimensionless numbers in capturing the full complexity of reactive transport, as they often neglect factors such as time-dependent phase transitions, local mixing heterogeneity, and inconsistent parameter definitions across studies (Algive et al., 2009; Mostaghimi et al., 2016; Ruzicka, 2008). Taken together, these findings highlight the need for a more nuanced, process-based understanding of $CaCO_3$ precipitation in complex fractures.

Calcium carbonate ($CaCO_3$) precipitation is a complex process involving nucleation, crystal growth, and phase transformation among polymorphs including amorphous calcium carbonate (ACC), vaterite, aragonite, and the thermodynamically stable calcite (Henzler et al., 2018; Jun et al., 2022; Noiriel et al., 2012; Voorhees, 1985). Nucleation, the initiation of the precipitation, is influenced by supersaturation levels, temperature, ion activity, and pH, as well as diffusion rate, critical radius, and interfacial energy (Li et al., 2014; Nooraiepour et al., 2021b). These parameters are central to classical nucleation theory (CNT), which describes nucleation as thermodynamically driven through the formation of a critical nucleus, while kinetic factors control the rate at which nucleation occurs (De Yoreo, 2003; Steefel and Van Cappellen, 1990). Furthermore, nonclassical nucleation mechanisms, such as pre-nucleation clusters or dense liquid intermediates, introduce additional complexity by making nucleation more probabilistic (Avaro et al., 2020; Gebauer et al., 2008; Jin et al., 2025; Nielsen et al., 2014). Random fluctuations and spatial heterogeneities, arising from molecular-scale thermal fluctuations and system-specific disorder, amplify this randomness, leading to variability in nucleation rates even under identical macroscopic conditions (Caupin and Grisenti, 2026; Goh et al., 2010). Following nucleation, the formation and growth of $CaCO_3$ polymorphs is heavily influenced by environmental conditions, including supersaturation, temperature, pH, and the presence of additives (Nielsen et al., 2014; San et al., 2023; Song et al., 2022). These factors control the formation and persistence of metastable phases like ACC, vaterite, and aragonite, which can transform into the more stable calcite. The kinetics of these transitions are crucial in determining the final mineralogy and crystal structure. While classical and stochastic models help explain above early-stage dynamics (Steefel and Van Cappellen, 1990), they struggle to account for the complexity introduced by nonclassical pathways and are remain limited by poorly constrained parameters, such as interfacial free energies and kinetic factors (Fernandez-Martinez et al., 2013; Li et al., 2014; Li and Jun 2018).

When flow is introduced into the system, it adds complexity by coupling nucleation and growth processes with hydrodynamic conditions. Under flow-through conditions, ACC often dominates early-stage precipitation at high supersaturation, likely via ultrafast nucleation through a dense liquid precursor (Morris et al., 2020), followed by transformation into vaterite, aragonite, or calcite (Morse et al., 2007). These phase transitions are strongly influenced by hydrodynamics. For example, higher flow rates enhance mixing and solute transport, which can promote the formation of vaterite by delaying its transformation into calcite (Han et al., 2005). Additionally, flow-driven transport in confined geometries has been shown to produce spatially distinct precipitation morphologies (Schuszter et al., 2016). Flow-through and microfluidic studies have significantly advanced our understanding of polymorph selection and transformation mechanisms in dynamic environments (Nisbet et al., 2025; Noiriel et al., 2012; Xu and Balhoff, 2023; Yoon et al., 2019). However, our ability to generalize $CaCO_3$ precipitation dynamics across systems remains limited, given the strong dependence on system-specific chemical and hydrodynamic conditions (Jun et al., 2022; Han et al., 2005; Zhang et al., 2010).

This system dependence not only limits cross-system generalization but also poses challenges for integrating experimental insights into predictive models (Steefel and Lasaga, 1994). While several studies have explored pore- to continuum-scale modelling approaches (Fazeli et al., 2020; Lassin et al., 2018; Li et al., 2017; Molins, 2015; Soulaine, 2024; Yang et al., 2022; Yoon et al., 2019; Zhang et al., 2022), developing robust frameworks remain difficult due to the inherent complexity of capturing how flow rate influences nucleation, growth, and phase transformation under in situ conditions. Additional factors—such as power-law particle size distributions driven by ripening dynamics (Xu and Balhoff, 2023) and dynamic porosity–permeability feedbacks linked to phase transitions (Molins et al., 2014; Wang et al., 2024; Weinhardt et al., 2022)—further complicate modelling efforts. These challenges are particularly pronounced in fractured media, where anisotropic transport and spatially variable reactivity generate evolving feedbacks between flow and precipitation (Steefel and Lichtner, 1994). In addition, recent microfluidic and numerical studies have begun to reveal how fracture-specific phenomena—such as fractured porous structure and inertial recirculating flows—shape solute transport and precipitation dynamics (Jiang et al., 2025; Steefel et al., 2005; Yang et al., 2024). These findings

highlight the need for integrative frameworks that can link microscale polymorph dynamics with macroscopic flow and transport in heterogeneous fractured systems.

In this study, we investigate how flow-induced transport dynamics regulate $CaCO_3$ precipitation processes within structurally heterogeneous fracture networks. Through microfluidic experiments, we explore how flow rates variations shape the spatial distribution of supersaturation and stochastic nucleation, govern the localization, persistence, and transformation of $CaCO_3$ polymorphs, and induce dynamic feedbacks between precipitation and evolving fracture permeability. Particular emphasis is placed on a hierarchical numerical strategy to examine the influence of heterogeneous flow and mixing on saturation development, and the probabilistic nucleation of minerals during the initial stage. This integrative approach provides a mechanistic basis for interpreting $CaCO_3$ precipitation dynamics in fractured media and offers new insights into the permeability evolution and reactive transport behavior relevant to subsurface carbon storage and other geochemical applications.

2. Materials and methods

2.1. Heterogeneous fracture structure design and micromodel fabrication

To investigate the impact of mixing on carbonate mineral reactions within heterogeneous fracture network, we designed and fabricated a micromodel (1.71 cm × 0.8 cm) that reproduced the hierarchical feature of natural fracture systems. The micromodel incorporated a dual-scale fracture network comprising macro-fractures (100–700 μm) and micro-fractures (80 ± 10 μm).

The design was inspired by a recent classification of natural fractures

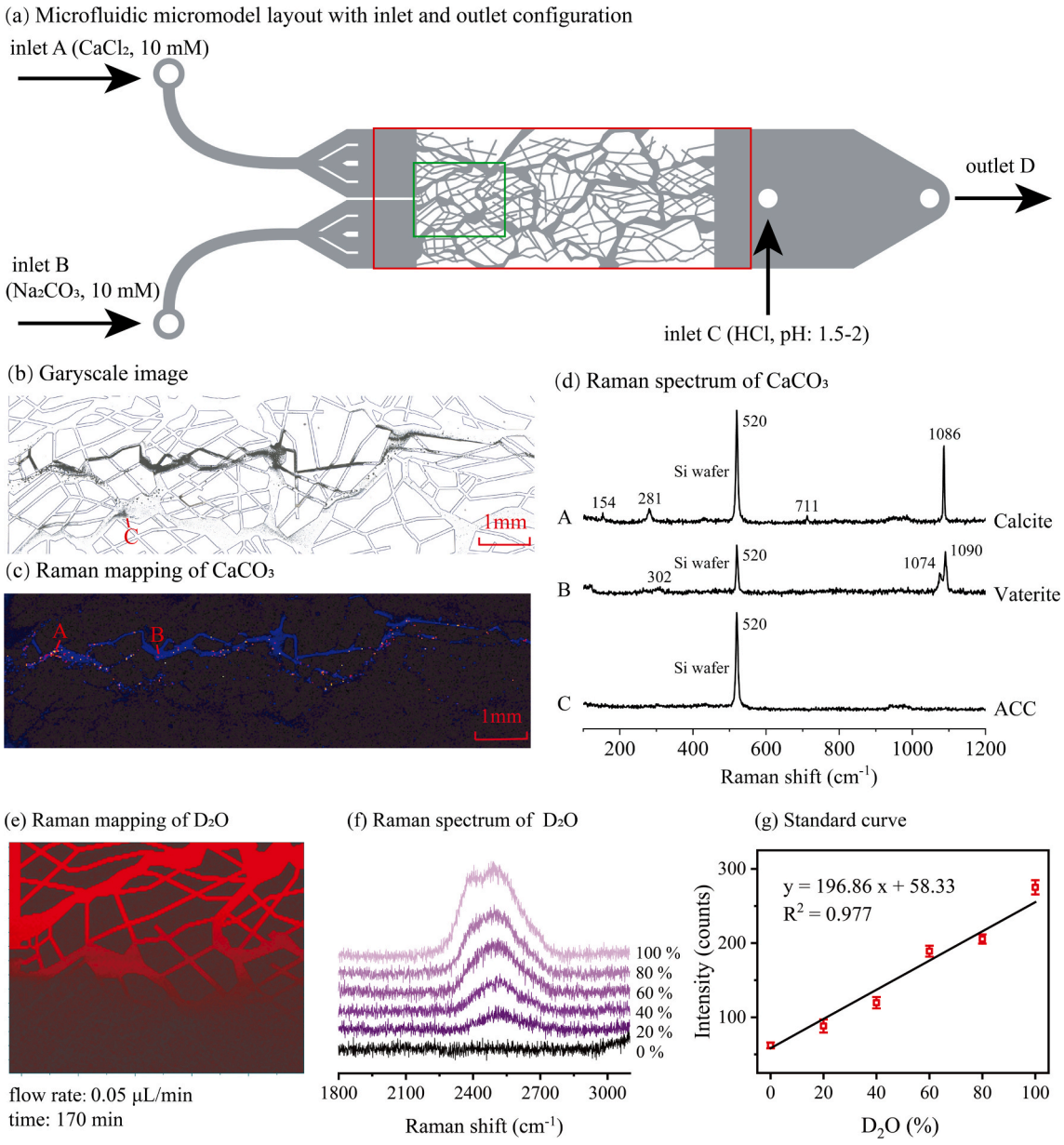

Fig. 1. Microfluidic micromodel setup and Raman spectroscopy for $CaCO_3$ precipitation identification and D_2O distribution. (a) Layout of the microfluidic micromodel, including inlets, outlets, and computational domain (red and green rectangles). (b) Grayscale image showing the fracture network and $CaCO_3$ precipitates. (c) Raman mapping image indicating the spatial distribution of calcite (red), vaterite/aragonite (red), and ACC (blue). (d) Representative Raman spectra of calcite (A), vaterite (B), and ACC (C) at the labeled locations in panels b and c. (e) Raman mapping of D_2O distribution. (f) Raman spectrum of D_2O with varying fractions. (g) Standard curve of D_2O fraction vs. Raman signal intensity.

into four aperture-based types (>100 μm, 40–100 μm, 20–40 μm, and 1–10 μm), where permeability increases by approximately one order of magnitude with each successive category (Mi et al., 2023). Type I (>100 μm) and Type II (40–100 μm) fractures were generally the most hydraulically active and well-connected, forming the main flow backbone and bridging pathways in reservoirs. To emulate this hierarchy, the fracture network was extracted from a scanning electron microscopy (SEM) image of a natural fracture surface in a carbonate rock core and served as the primary conductive channels. Within the fracture network, macro-fractures were mechanically induced fractures formed under stress localization, reflecting natural fracture patterns that resulted from the deformation process, while micro-fractures were inherited from the mineralogical fabric of the rock, representing the material's inherent anisotropy. The orientations of these micro-fractures, as derived from SEM images, reflected the rock's anisotropic nature. The network geometry was refined in AutoCAD by smoothing the edges of the macro-fractures, adjusting aperture sizes, and intentionally increasing the micro-fracture density to a value higher than typical real-world Type II fracture densities. This higher fracture density was chosen to enhance connectivity with the macro-fractures, facilitating flow and transport, and to aid in experimental observation and computational efficiency, which is a common design choice in microfluidic experimental setups (Godoy et al., 2025; Morais et al., 2020; Roman et al., 2016) (Fig. 1a).

For micromodel fabrication, two inlet channels (A and B) were directly connected to the fracture network, while an additional inlet (C) and outlet (D) were positioned downstream to allow acid injection and minimize clogging (Zhang et al., 2010). The fracture network was etched to a depth of 19–21 μm in a silicon wafer using standard photolithographic patterning and deep reactive ion etching (Chomsurin and Werth, 2003; Liu et al., 2012). The wafer was then oxidized at 1100 °C for 1.5 h to form a ~ 150 nm SiO_2 layer and anodically bonded to borosilicate glass (BF33) to seal the flow channels.

2.2. Micromodel experiments

The micromodel was mounted on a house-made manifold and assembled onto the automated stage of a confocal microscopic Raman spectrometer (LabRAM HR Evolution, HORIBA). Before each experiment, the micromodel was flushed with CO_2 gas, followed by the injection of more than 100 pore volumes (PV) of deionized (DI) water to ensure thorough saturation of the fracture network. Subsequently, 10 mM $CaCl_2$ (>97%, Sigma-Aldrich) and 10 mM Na_2CO_3 (>99.5%, Sigma-Aldrich) were simultaneously injected into the micromodel through inlets A and B, respectively, at equal flow rates using a dual-channel syringe pump (Harvard Apparatus, Model Pump 11 Pico Plus Elite) at room temperature (25 ± 2℃). The experiments were conducted at volumetric flow rates of 5, 0.5, and 0.05 μL/min for each of the inlet A and B. To prevent the risk of $CaCO_3$ precipitation and potential clogging at the outlet, an HCl solution (pH 1.5–2.0) was concurrently injected through inlet C at the same flow rate used for inlets A and B, using another syringe pump (SP-01, ShenChen).

To ensure the establishment of a stable flow field before the introduction of Ca^{2+} and CO_3^{2-} ions, the syringes were connected to the micromodel via tubing with an internal diameter of 1/150 in. and a length of 40 cm. This configuration provided stabilization times of 1.8, 18, and 180 min at flow rates of 5, 0.5, and 0.05 μL/min, respectively, which were sufficient to establish a stable flow field in the micromodel before the injection of the reactants.

The process of $CaCO_3$ precipitation and its dynamic mineralogical transformations were monitored using the confocal microscopic Raman spectrometer integrated with optical imaging. At selected time intervals, wide-field visualization of the entire micromodel was achieved by capturing 120 overlapping images with a CCD camera through a 5 × objective lens, arranged in a 15 × 8 grid. These images were subsequently montaged into a single composite image using LabSpec6 software (Fig. 1b). To track the mineralization process in greater detail, high-resolution snapshots were acquired at specific locations within the fracture network using a 10 × objective lens. For mineral phase identification, swift Raman spectroscopy mapping was performed using a 100 mW, 532 nm laser with an acquisition time of 0.1 s. Spectra were collected at 30 μm intervals across areas exhibiting $CaCO_3$ precipitation (Fig. 1c), covering the spectral range from 100 to 1300 cm^{-1} using a 20 × objective lens.

A separate set of experiment was performed to investigate variations in fluid mixing during the precipitation process using a deuterium oxide (D_2O; >97%, Sigma-Aldrich) tracer. Prior to each experiment, the fracture network was saturated with DI water. Subsequently, 10 mM $CaCl_2$ in D_2O and 10 mM Na_2CO_3 in DI water were simultaneously injected through inlets A and B, following the same flow conditions used in the precipitation experiments. Mineral precipitation and fluid mixing were monitored at two selected injected pore volumes (PV): 20 PV and 400 PV, both near the injection points (Fig. 1e). This limited spatial focus was chosen to allow detailed monitoring of fluid mixing dynamics, as mixing evolves rapidly while Raman mapping requires more time to complete. These injected PV values represent two distinct stages: the initial stage prior to $CaCO_3$ precipitation begins and the extended stage after calcite nucleation has slowed or ceased (detailed in section 3). The mineral precipitation bands were recorded through wide-field optical imaging, using 42 overlapping sub-images (6 × 7 grid). For fluid mixing analysis, rapid Raman mapping was performed using a 532 nm, 100 mW laser with an acquisition time of 0.15 s per spectrum. Raman spectra, covering the spectral range from 1800 to 3100 cm^{-1}, were collected at 30 μm spatial intervals across the inlet mixing region under a 50 × objective lens.

2.3. Quantification of $CaCO_3$ and its mineral phases

$CaCO_3$ precipitation was quantified by grayscale image analysis and its mineralogy identified by Raman mapping. Montaged grayscale images were processed in Image J by cropping extraneous borders, converting to 8-bit, and subtracting the fracture network background using the Image Calculator function. The processed images were then binarized to isolate $CaCO_3$ precipitates. Mineral mass was estimated as:

$$M = \frac{A \times d}{V_{mol}} \tag{1}$$

where A is the area of precipitates (number of pixels × pixel area, 31.74 μm^2), d is the micromodel depth, and V_{mol} is the molar volume of the mineral (36.93 cm^3/mol for calcite and 37.47 cm^3/mol for metastable phases) (Noiriel et al., 2012; Poonoosamy et al., 2019; Sulpis et al., 2022; Yoon et al., 2012).

Mineralogical identification was performed by aligning Raman mapping images with the grayscale images using a custom MATLAB code (Fig. 1b and c). To ensure pixel-by-pixel comparison, the resolution of grayscale images (3040 × 1420 pixels) was downscaled to match the Raman maps (573 × 265 pixels). Calcite, vaterite, and aragonite were identified based on their characteristic Raman spectral signatures (Boyd et al., 2014) (Fig. 1d), while pixels visible in grayscale but lacking diagnostic Raman signals were classified as amorphous calcium carbonate (ACC) (Fig. 1b and c). Due to the weak or overlapping peaks of vaterite and aragonite, distinguishing between these phases in Raman spectroscopy is challenging. To simplify quantification, we grouped the $CaCO_3$ precipitates into two categories: calcite and metastable $CaCO_3$ (ACC, vaterite, and aragonite). This classification is based on the fact that all metastable $CaCO_3$ phases are thermodynamically unstable relative to calcite and tend to transform into calcite over time. By combining these metastable phases, we not only address the difficulties in distinguishing vaterite from aragonite but also streamline the quantification process, ensuring it still reflects the key dynamics of $CaCO_3$ precipitation.

2.4. Quantification of D_2O distribution and mixing degree

The spatial distribution of D_2O was quantified using a standard curve correlating Raman intensity with D_2O volume fraction. Standard samples were prepared by mixing H_2O and D_2O at volume fractions ranging from 0% to 100% D_2O. Aliquots of each standard were deposited onto glass slides to form thin water films and analyzed by Raman spectroscopy under the same conditions as the in-situ measurements for Raman mapping in microfluidic experiments (Fig. 1f). Eleven Raman spectra were collected per standard to assess reproducibility. The 2200–2800 cm^{-1} region, characteristic of the O–D stretching vibrations, shows a prominent peak directly proportional to the D_2O concentration (Lönartz et al., 2023). The maximum peak intensity within this region is commonly used to establish a quantitative relationship between the Raman signal and the D_2O fraction in mixed H_2O/D_2O solutions (Lönartz et al., 2023) (Fig. 1g. This relationship was applied to Raman mapping data (Fig. 1e) to retrieve the spatial distribution of D_2O. The Raman maps consisted of 134 × 172 pixels, each representing local D_2O concentration.

Since the aqueous phase in the mixing zone consists only of H_2O and D_2O, and their total volume fraction equals unity, the volumetric concentration of H_2O was calculated as:

$$C_{H_20} = 1 - C_{D_2O} \tag{2}$$

Given that the solutions injected through inlets A and B were 100% D_2O and 100% H_2O, respectively, the local mixing intensity was quantified as the product of the two volume fractions, $C_{D_20}(x,y) \times C_{H_20}(x,y)$, which reaches its maximum when both components are present in equal proportions. All Raman mapping data analyzing procedures, including peak extraction, volume fraction conversion, and mixing intensity calculation, were implemented using custom MATLAB codes.

2.5. Numerical modeling

A fully coupled, pore-scale simulation that explicitly resolves flow dynamics, chemical mixing, stochastic nucleation, mineral growth and transformation, and their feedback on fracture geometry remains extremely challenging. Such an approach would require introducing numerous poorly constrained parameters across multiple spatial and temporal scales, which can obscure mechanistic interpretation when the objective is to isolate the controlling processes governing $CaCO_3$ precipitation. Moreover, achieving the spatial resolution necessary to explicitly represent reactive nucleation and crystal growth within our fracture geometry would result in large computational demands, rendering such fully coupled pore-scale simulations impractical for systematic investigation.

To address this challenge, we adopted a hierarchical numerical strategy consisting of two pore-scale simulations and one Darcy-scale simulation, each designed to interrogate a specific component of the coupled system. First, pore-scale fluid dynamic simulations were performed to resolve the flow field within the heterogeneous fracture geometry. Second, these velocity fields were used to conduct pore-scale solute transport simulations at the initial stage to quantify how flow-controlled mixing determines the spatial distribution of supersaturation. Finally, a Darcy-scale reactive transport simulation was employed to examine how flow-controlled mixing governs the spatial variability of nucleation propensity, independent of geometric feedbacks from precipitation. In the final simulation, a Darcy-scale framework was deliberately adopted to isolate the intrinsic control of flow-driven mixing on nucleation propensity by suppressing feedbacks arising from nucleation- and precipitation-induced fracture geometry evolution. This multi-level modeling framework allows the effects of flow, mixing, and nucleation to be evaluated in a controlled and complementary manner, providing mechanistic insight into the role of transport-driven heterogeneity in regulating $CaCO_3$ precipitation pathways.

Pore scale numerical simulations were performed using COMSOL Multiphysics 6.0 to investigate fluid dynamics in the 2-D heterogeneous fracture network. The computational domain (8 mm × 22.12 mm) comprised the entire fracture geometry, an upstream inlet section (8 mm × 3 mm), and a downstream outlet section (8 mm × 2 mm) (Fig. 1a, red rectangle). The Reynolds numbers (Re) corresponding to the experimental flow rates (5, 0.5, and 0.05 μL/min) were 0.225, 0.0225, and 0.00225, using a characteristic length scale of 20 μm, the depth of the microfluidic channels. These Re values confirmed laminar conditions. Accordingly, steady-state velocity fields were obtained by solving the Navier-Stokes equations with the finite element method (FEM) (Zeng et al., 2018):

$$\nabla \bullet (\boldsymbol{uu}) - \nabla \bullet (\mu \nabla \boldsymbol{u}) = -\nabla p \tag{3}$$

$$\nabla \bullet \boldsymbol{u} = 0 \tag{4}$$

where $\boldsymbol{u}$ is the velocity vector (m/s), μ is the dynamic viscosity of water (Pa·s), and p is the pressure (Pa). Zero initial velocity and pressure were specified throughout the domain. Fixed pressure boundary conditions were imposed at the inlets and calibrated to reproduce the experimental volumetric flow rates (5, 0.5, and 0.05 μL/min), while the outlet pressure was set to zero. No-slip boundary conditions were applied on all solid surfaces, including the fracture walls and, where present, pre-existing $CaCO_3$ surfaces.

Simulations were conducted for two representative stages: the initial stage and the extended stage. During the extended stage, existing calcite crystals continue to grow, while metastable $CaCO_3$ phases increase in both nucleation frequency and crystal size (as detailed in Section 3). To evaluate the hydrodynamic consequences of precipitation, $CaCO_3$ formed during this stage was incorporated into the model as a static solid phase, enabling isolation of precipitation-induced geometric effects on flow redistribution. For the extended-stage, simulation times were set to 1 h for flow rate of 5 μL/min and 19 h for 0.05 μL/min. These times were selected to yield comparable total precipitated amounts (~0.1 μmol) across flow rates, while preserving differences in spatial distribution arising from distinct prior flow–reaction histories.

The steady-state velocity fields obtained for the initial stages prior to $CaCO_3$ precipitates were used to compute solute transport within the fracture network. At this stage, precipitation reactions were excluded to isolate the influence of transport on solute mixing and the distribution of supersaturation. This simplification is supported by experimental observations showing negligible $CaCO_3$ precipitation within the first ~ 20 PV (as detailed in Section 3), which is also sufficient to characterize solute migration across the entire fracture domain. Solute transport was solved using the advection–diffusion equations (Tournassat and Steefel, 2019; Zeng et al., 2018):

$$\frac{\partial \Psi_i}{\partial t} = D_i \nabla^2 \Psi_i - \boldsymbol{u} \bullet \nabla \Psi_i \tag{5}$$

where Ψ_i is aqueous component concentration (mol/m^3) and D_i is its diffusion coefficient (m^2/s), both of which are listed in Table 1. Conservative aqueous components (Ca^{2+}, Na^+, and Cl^-) and total inorganic carbon (DIC) were transported explicitly using COMSOL Multiphysics. Initial concentrations within the fracture network were set to zero. At the inlets, Dirichlet boundary conditions were prescribed based on experimental concentrations, while a Neumann boundary condition (zero diffusive flux) was applied at the outlet. Zero-flux boundary

Table 1
Parameters used in simulations.

Parameter	Value	Unit	Reference
D_{Na^+}	1.33×10^{-9}	m^2/s	(Haynes, 2014)
$D_{Ca^{2+}}$	7.92×10^{-9}	m^2/s	(Haynes, 2014)
D_{DIC}	9.23×10^{-10}	m^2/s	(Haynes, 2014)
D_{Cl^-}	2.03×10^{-10}	m^2/s	(Haynes, 2014)

conditions were imposed on the fracture walls and $CaCO_3$ surfaces.

Following the transport simulations, carbonate speciation (H_2CO_3, HCO_3^-, and CO_3^{2-}), activity coefficients, and pH were determined using local aqueous equilibrium calculations with the IPhreeqc geochemical module (IPhreeqcCOM 3.7.3) and the MINTEQ thermodynamic database, assuming that acid-base reactions equilibrate rapidly relative to solute transport. Custom MATLAB scripts were used to interface COMSOL transport outputs with IPhreeqc equilibrium calculations and map the resulting carbonate speciation and ion activity products (IAP = $\alpha_{Ca^{2+}}\alpha_{CO_3^{2-}}$) of $CaCO_3$ onto the computational grid.

Darcy-scale reactive transport simulations were performed using CrunchTope to investigate the influence of flow-controlled mixing on the spatial variability of $CaCO_3$ nucleation during the initial stage (< 1 h), during which nucleation is assumed to have no significant effect on porosity or transport properties (Steefel et al., 2015; Steefel and Van Cappellen, 1990). This assumption is supported by experimental observations indicating negligible changes in fracture geometry during the initial stage (as detailed in Section 3). Since the primary focus was on nucleation dynamics, the crystal growth process was not considered here, as it occurs on a longer timescale compared to the initial nucleation phase. Moreover, in CrunchTope, both nucleation and crystal growth were considered as part of the final output, but their combined effects made it difficult to independently distinguish the nucleation rate. Therefore, to isolate the contribution of nucleation to the overall mineralization process, we focused solely on the nucleation stage in our simulations. The governing equation is expressed as:

$$\frac{\partial \phi \Psi_i}{\partial t} = \nabla \bullet (\phi D_m \nabla \Psi_i) - \nabla \bullet (\boldsymbol{\nu} \phi \Psi_i) - \gamma_i R_{i,nuc} \tag{6}$$

where ϕ is porosity, Ψ_i is aqueous component concentration (mol/m^3), D_m is the molecular diffusion coefficient for all components (1.0×10^{-9} m^2/s), $\boldsymbol{\nu}$ is velocity (m/s) in the Darcy-scale simulations, and γ_i is the stoichiometric coefficient. The transport was solved for the total concentration of the aqueous components, while their speciation was determined by thermodynamic equilibrium constraints to calculate nucleation rate $R_{i,nuc}$ (mol/m^3•s). $R_{i,nuc}$ was modeled using classical nucleation theory implemented in CrunchTope, as follows (Deng et al., 2021; Li et al., 2017; Steefel et al., 2015):

$$R_{i,nuc} = A_0 J_0 \frac{\tau_N}{\tau_p} exp\left[-\frac{16\pi V^2 \alpha^3}{3k_B^3 T^3 \left(\ln\left(\frac{IAP}{K_{sp}}\right)\right)^2} \right] \tag{7}$$

where A_0 is set to unity (m^2 /m^3), J_0 is the kinetic factor with a value of 1.0×10^{-8}mol/m^2·s for calcite (Li et al., 2017), τ_N is the theoretical nucleation time following the CNT, τ_p is a stochastic induction time, V is the molecular volume (6.13×10^{-29}m^3 for calcite), α is the effective interfacial tension for calcite with a value of 0.047 J/m^2 (Li et al., 2017), k_B is the Boltzmann constant (1.38×10^{-23} J/K), T is temperature (298 K), and K_{sp} is solubility product of calcite (3.3113×10^{-9}) (Plummer and Busenberg, 1982). Here, calcite was chosen as the primary mineral for nucleation because it is the thermodynamically stable phase in the $CaCO_3$ system and the most likely to form in the initial stage under the experimental conditions (as detailed in Section 3). To prevent simultaneous nucleation across all computational elements in continuous mass models, random fluctuations in nucleation times were incorporated in CrunchTope based on CNT by applying a random perturbation to the theoretical nucleation time τ_N within each grid cell (Deng et al., 2022). This perturbation follows a standard normal distribution:

$$\tau_p = \frac{x + n\sigma}{n\sigma} \tau_N \tag{8}$$

where x is a random variable drawn from $N(0, 1)$, σ is the standard deviation of the normal distribution and equals to 1 in our case, and n = 4, capturing 99.9% of the population in the normal distribution (Deng et al., 2022; Fazeli et al., 2020; Nooraiepour et al., 2021a). This approach allows nucleation events to occur at different times within the computational domain, introducing spatial variability and randomness into the nucleation process.

A localized inlet mixing region (4.02 mm × 5.16 mm) was extracted for simulation under experimental flow rates (5, 0.5, and 0.05 μL/min). This domain corresponds to the region analyzed in the D_2O–H_2O mixing experiments (green rectangle in Fig. 1a). Porosity (0.3) was assigned based on the fracture network's volume fraction. The velocity and concentration fields within the entire computational domain were initialized to zero. Then, fluid transport was implemented using the pump formulation in CrunchTope, with pump rates distributed across 340 inlet points to replicate the experimental inlet flow conditions. The upper and lower boundaries were defined as impermeable. At inlet A and B, the aqueous component concentrations were fixed to those of the experimental solutions.

3. Results and discussion

3.1. $CaCO_3$ precipitation at different flow rates

To understand the dynamics of $CaCO_3$ precipitation within the heterogeneous fracture network, we examined its spatial and temporal variations under flow rates of 5, 0.5, and 0.05 μL/min (Figs. 2, 3a, and 3b). At the early precipitation period, $CaCO_3$ primarily accumulated in the central mixing zone (Fig. 2). Quantitative measurement over time showed that the amount of $CaCO_3$ increased with flow rates, suggesting that faster delivery of Ca^{2+} and CO_3^{2+} at higher flow rates promoted greater precipitation (Fig. 3a2). However, when normalized by pore volume (PV), lower flow rates yielded more $CaCO_3$ at the same PV, indicating that longer residence time allowed more complete precipitation (Fig. 3b2). While this trend was consistent during the early precipitation period, a distinct shift in behavior emerged after approximately 100 PV. At the highest flow rate (5 μL/min), precipitation patterns remained relatively uniform across experiments (Exp. 1 and 2), with the precipitation rate declining sharply after 2 to 4 h (700–1400 PV). This decline likely resulted from rapid $CaCO_3$ accumulation that clogged the mixing zone, limiting ion supply and confining precipitation to a narrow zone (Fig. 2). In contrast, at lower flow rates (0.5 and 0.05 μL/min), the evolution of precipitation patterns became highly variable. In some experiments (Exp. 3 and 5), precipitation zones continued to expand, leading to sustained accumulation of $CaCO_3$. However, in other experiments (Exp. 4 and 6), localized precipitation caused rapid clogging beyond 100 PV, significantly reducing deposition rates (Fig. 2).

The deposition of $CaCO_3$ within heterogeneous fractures shares key characteristics with phenomena observed in pore networks—particularly the formation of a central mixing zone (Boyd et al., 2014; Yoon et al., 2019)—but is inherently more complex. In pore networks, flow rate is a primary factor governing the final extend of $CaCO_3$ precipitation, with slower flow generally leading to greater mineral accumulation due to longer fluid-rock interaction times (Algive et al., 2009). However, in our fracture systems, although flow rate dominated precipitation behavior during the first few hours, the continued evolution of $CaCO_3$ precipitation led to considerable divergence after 100 PV. Beyond this point, significant variation in both fracture structure evolution and the final $CaCO_3$ precipitation trends and amounts was observed, even at the same flow rate. The relationship between flow rate and $CaCO_3$ precipitation became increasing complex and difficult to predict. For example, at 0.05 μL/min, one experiment (Exp. 6) resulted in a greater amount of $CaCO_3$ precipitation compared to an experiment conducted at 0.5 μL/min (Exp. 4) at the same PV, while another experiment at 0.05 μL/min (Exp. 5) produced less precipitation than Exp. 4. Interestingly, at lower flow rates, the variation in $CaCO_3$ precipitation was much more pronounced, even though the flow rate remained constant. This underscores how evolving fracture geometry and flow dynamics increasingly

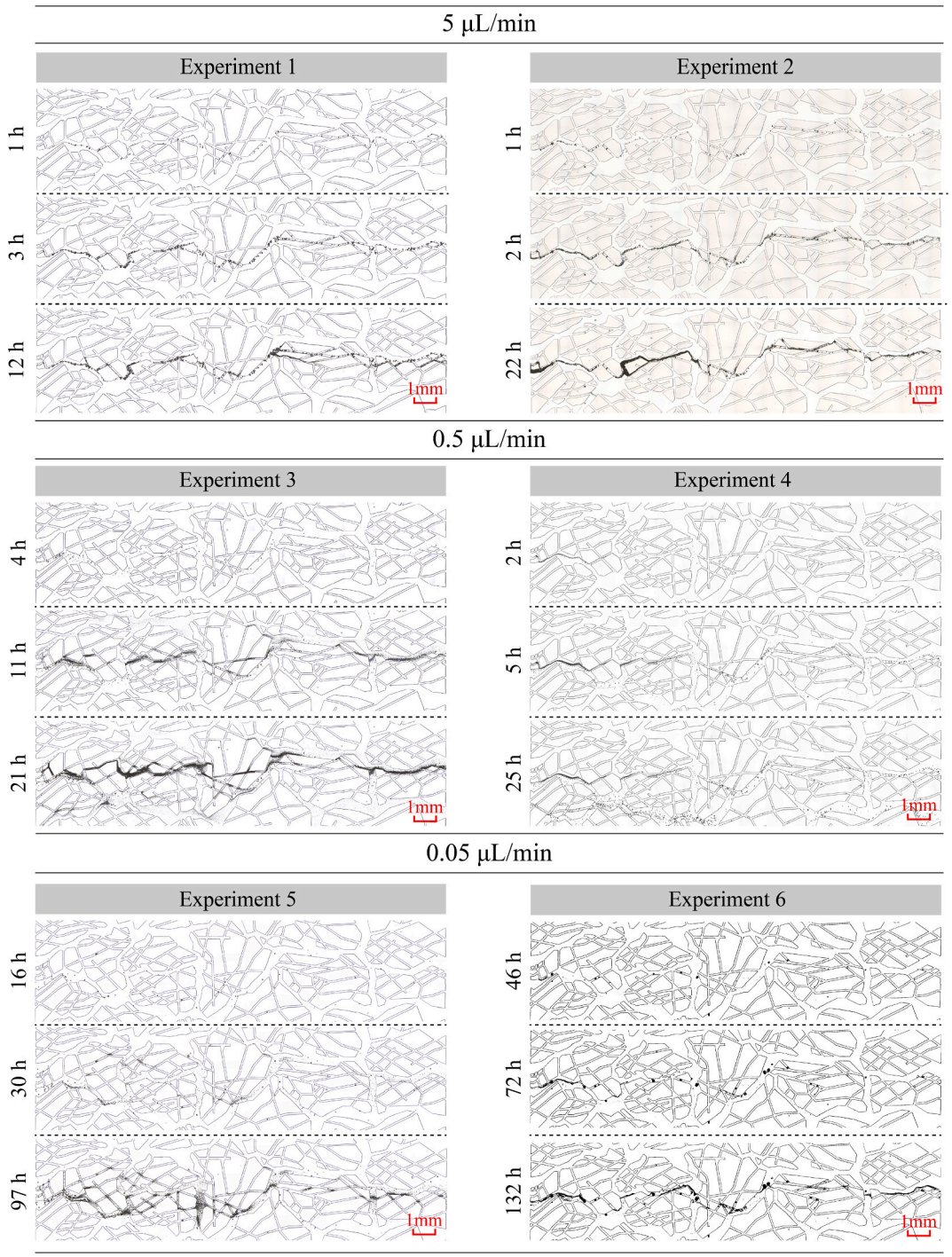

Fig. 2. Temporal evolution of $CaCO_3$ precipitation at various flow rates in heterogeneous fracture network.

influence the distribution of $CaCO_3$ precipitation, highlighting the complex relationship between flow rate and mineral deposition.

3.2. *Influence of $CaCO_3$ mineral phases on precipitation and fracture evolution at different flow rates*

To investigate the impact of $CaCO_3$ precipitation on fracture evolution within a heterogeneous fracture network, microscopic imaging and Raman spectroscopy were used to characterize crystallization processes and mineral phases (Fig. 4) in two distinct domains along the mixing zone under varying flow rates (Fig. 5a). In domain A, near the inlet, streamline-shaped ACC formed rapidly in localized regions and transformed into more stable phases such as calcite at flow rates of 5 and 0.5 μL/min (Fig. 5b). However, at 0.5 μL/min in most areas of domain A (Fig. 5b), and throughout domain A at 0.05 μL/min (not shown), calcite formed directly. In domain B, located downstream, calcite was the dominant phase from the onset across all flow rates, forming as larger crystals. Over time, the nucleation of new calcite particles slowed or ceased, while existing calcite crystals continued to grow in size. Simultaneously, metastable phases, including ACC, vaterite, and occasionally aragonite, appeared between growing calcite crystals. These metastable phases grew until filling interstitial spaces, ultimately causing blockage (Fig. 5b).

The temporal evolution of $CaCO_3$ mineral phases across the entire

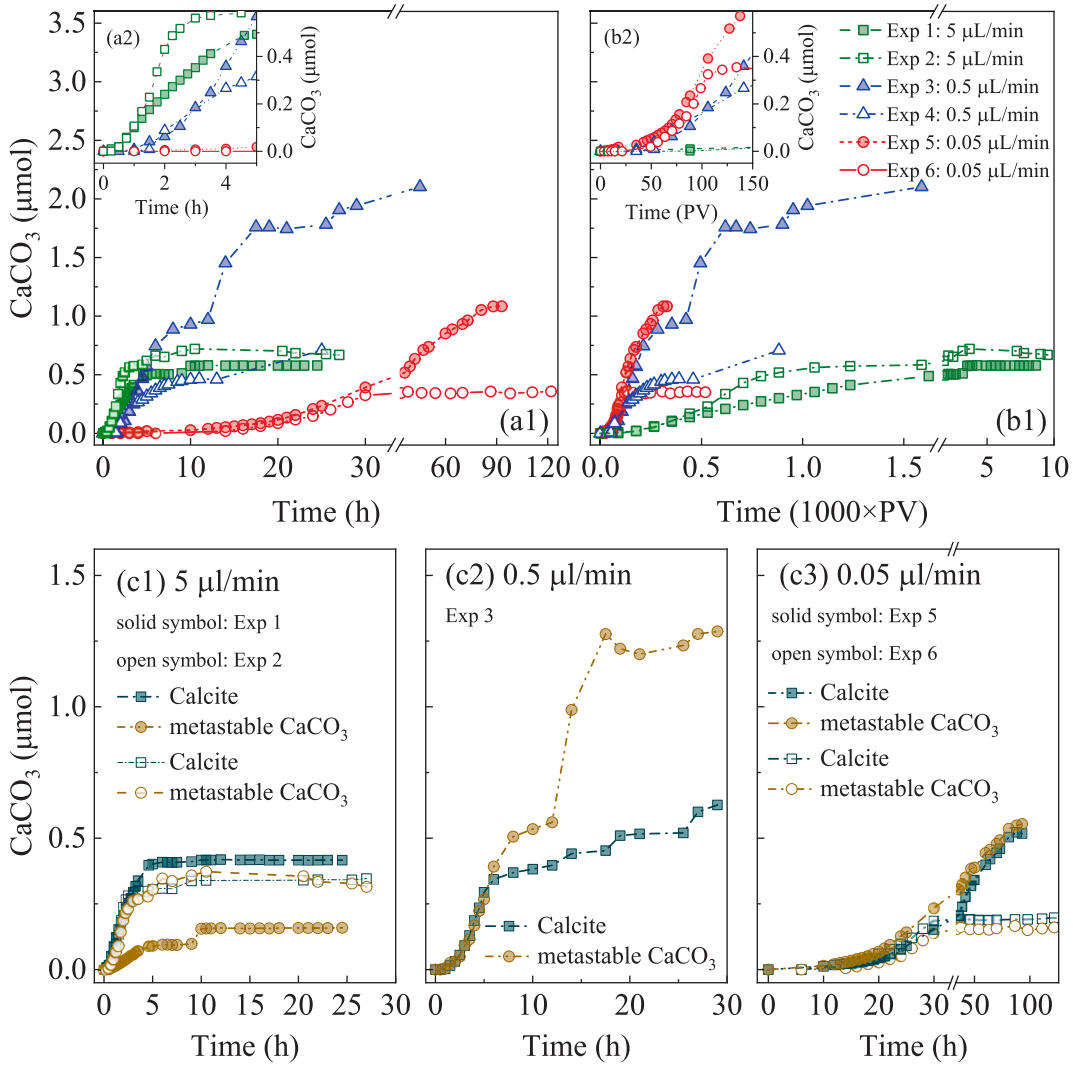

Fig. 3. Temporal evolution of $CaCO_3$ precipitation at different flow rates (5, 0.5, and 0.05 μL/min) in a heterogenous fracture micromodel, illustrating the formation of $CaCO_3$ over time (a) and pore volume (PV) (b). Panels c1-c3 show the progression of calcite and metastable $CaCO_3$ phases (ACC, vaterite, and aragonite) as a function of time for different flow rates. Each flow rate includes results from two experimental runs, showing the variability in mineral formation. Exp 4 lacks Raman spectroscopy data; so, the $CaCO_3$ mineral phases in panel c2 do not include Exp 4.

micromodel under different flow rates is presented in Fig. 3c. The relative abundance of calcite and metastable $CaCO_3$ phases varies significantly. For example, when calcite nucleated was denser (Exp. 1 in Fig. 3c1 and Fig. 5b, 5 μL/min), limited space was available for the growth of metastable phases, leading to faster blockage of the fracture network and reduced precipitation of metastable $CaCO_3$ phases over time. In contrast, when calcite nucleation was sparser (Exp. 3 in Fig. 3c2 and Fig. 5b, 0.5 μL/min), more space remained for metastable phases to grow, allowing continued precipitation throughout the experiment.

This difference is linked to the local deposition patterns of each phase. Calcite forms dense rhombohedral crystals, while metastable phases such as ACC, vaterite, and aragonite exhibit porous crystal morphologies that affect their packing and permeability (Noiriel et al., 2016; Weinhardt et al., 2021). ACC forms amorphous nanoparticle clusters, vaterite forms needle-like polyhedral aggregates, and aragonite forms elongated, needle-like or prismatic structures. These structural differences significantly influence ion transport. The more compact calcite blocks flow and precipitation once nucleation sites are occupied, while the less-dense metastable phases, with their porous structures, facilitate sustained ion transport and further precipitation.

These results underscore the pivotal role of $CaCO_3$ polymorphs in influencing fracture geometry and controlling final precipitation outcomes via their interplay with mass transfer processes. The formation and phase transformation of $CaCO_3$ polymorphs are modulated by local saturation conditions and the kinetics of nucleation and growth—both strongly affected by transport conditions. This dynamic feedback between evolving mineral structures and transport pathways can further reinforce spatial heterogeneity in precipitation.

3.3. *Flow-controlled mixing and $CaCO_3$ precipitation with hydrodynamic feedbacks*

Given that flow rate is a key control on solute transport, we examine how variations in flow rate regulate D_2O–H_2O mixing in both the absence and presence of $CaCO_3$ precipitation (Fig. 6). At the initial stage (20 PV), before precipitation occurred, mixing was governed by the balance between advection and transverse diffusion. Higher flow rates produced a narrow mixing band, whereas decreasing flow rate progressively broadened the mixing zone due to enhanced transverse diffusion (Fig. 6a1–c1). These observations establish the baseline mixing geometry upon which subsequent $CaCO_3$ precipitation develops. As precipitation developed, $CaCO_3$ preferentially formed along the mixing band, with a systematic shift toward the CO_3^{2-}-rich side, consistent with the higher diffusion coefficient of Ca^{2+} relative to CO_3^{2-} (Fig. 6a2–c2). At

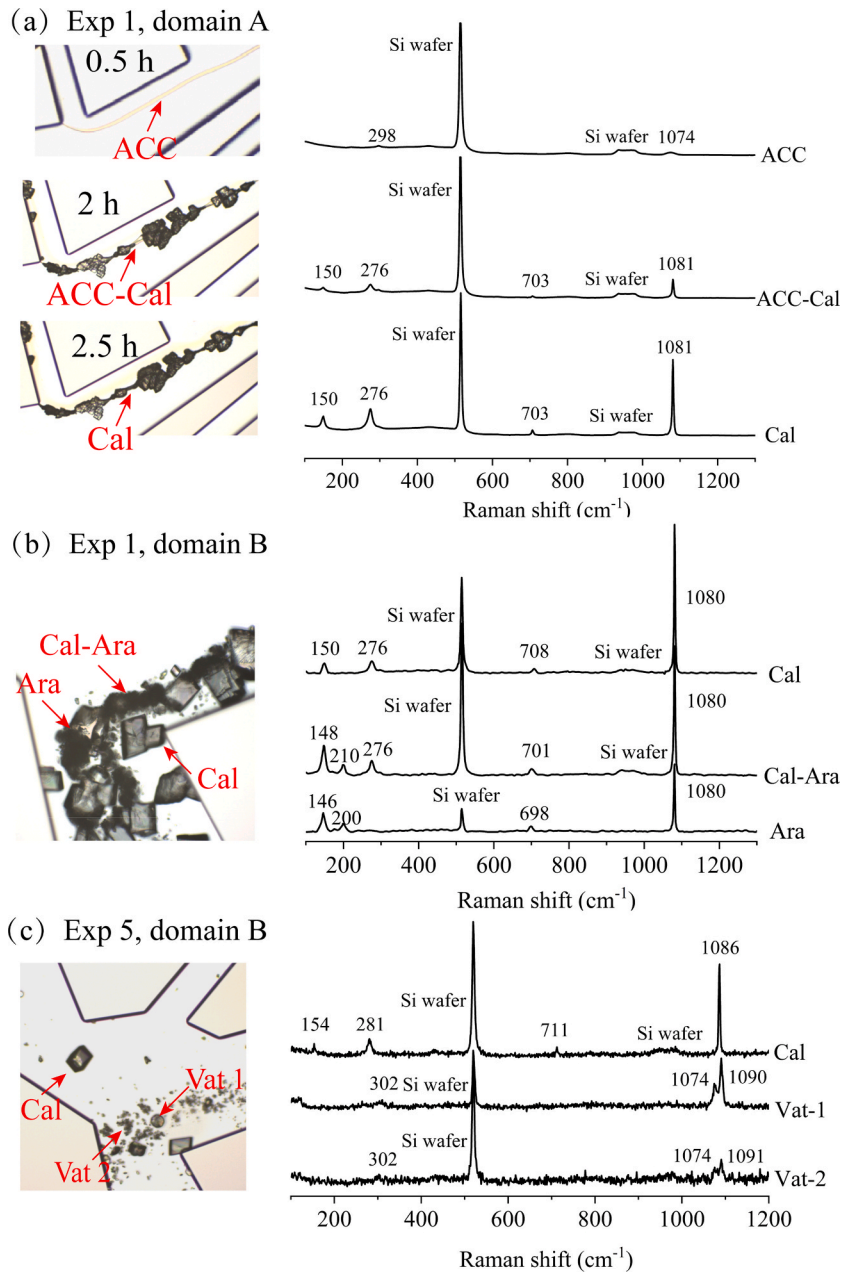

Fig. 4. Morphology and Raman spectra of $CaCO_3$ minerals. (a) Domain A (inlet) in experiment 1 at 5 μL/min. (b) Domain B (downstream) in experiment 1 at 5 μL/min. (c) Domain B (downstream) in experiment 5 at 0.05 μL/min. The location of Domain A and B are shown in Fig. 5a.

the extended stage (400 PV), accumulated precipitation formed spatially heterogeneous partial blockage structures within preferential flow paths, redistributing flow and consequently narrowing the effective mixing band and reducing mixing intensity relative to the initial state (Fig. 6a1–c1 vs. Fig. 6a3–c3).

To quantify how precipitation-induced heterogeneity feeds back on hydrodynamics, we computed flow field distributions at both the initial stage (prior to calcite precipitation) and the extended stages (after calcite nucleation slows or ceases) for Exp. 1 and Exp. 2 (5 μL/min) and Exp. 5 (0.05 μL/min) (Fig. 7). At 5 μL/min, substantial $CaCO_3$ precipitation developed along the dominant mixing band. As precipitation accumulated, a pronounced reduction in local flow velocity emerged within this region after 1 h of injection, particularly in upstream regions (Fig. 7b and c). The relative roles of advection and diffusion were quantified by calculating spatially averaged velocities within domain A and domain B and estimating the local Peclet number in the mixing zone ($Pe = UL/D_m$), using a characteristic length scale of 20 μm and a molecular diffusion coefficient of 1×10^{-9} m^2/s. At the initial stage, the local Pe number in domain A was 71.2. After 1 h, the Pe number decreased to 36.0 in Exp. 1 and 31.1 in Exp. 2. These values indicate that solute transport and mixing were predominantly advection-controlled under high-flow conditions during most of the experimental duration, with progressive transition toward diffusion-controlled transport as clogging evolved. The stochastic spatial distribution of precipitation sites generated heterogeneous blockage structures, which locally redistributed flow and produced distinct velocity patterns even among replicate experiments conducted at the same flow rate (Fig. 7b2 vs c2 and b3 vs c3). At 0.05 μL/min, reductions in local velocity were also observed, although $CaCO_3$ precipitation was more spatially sparse (Fig. 7e). In domain A, the initial local Pe number was 0.71 and decreased to 0.5 during the extended stage. Compared to the advection-dominated regime at high flow rates, these values reflect a diffusion-

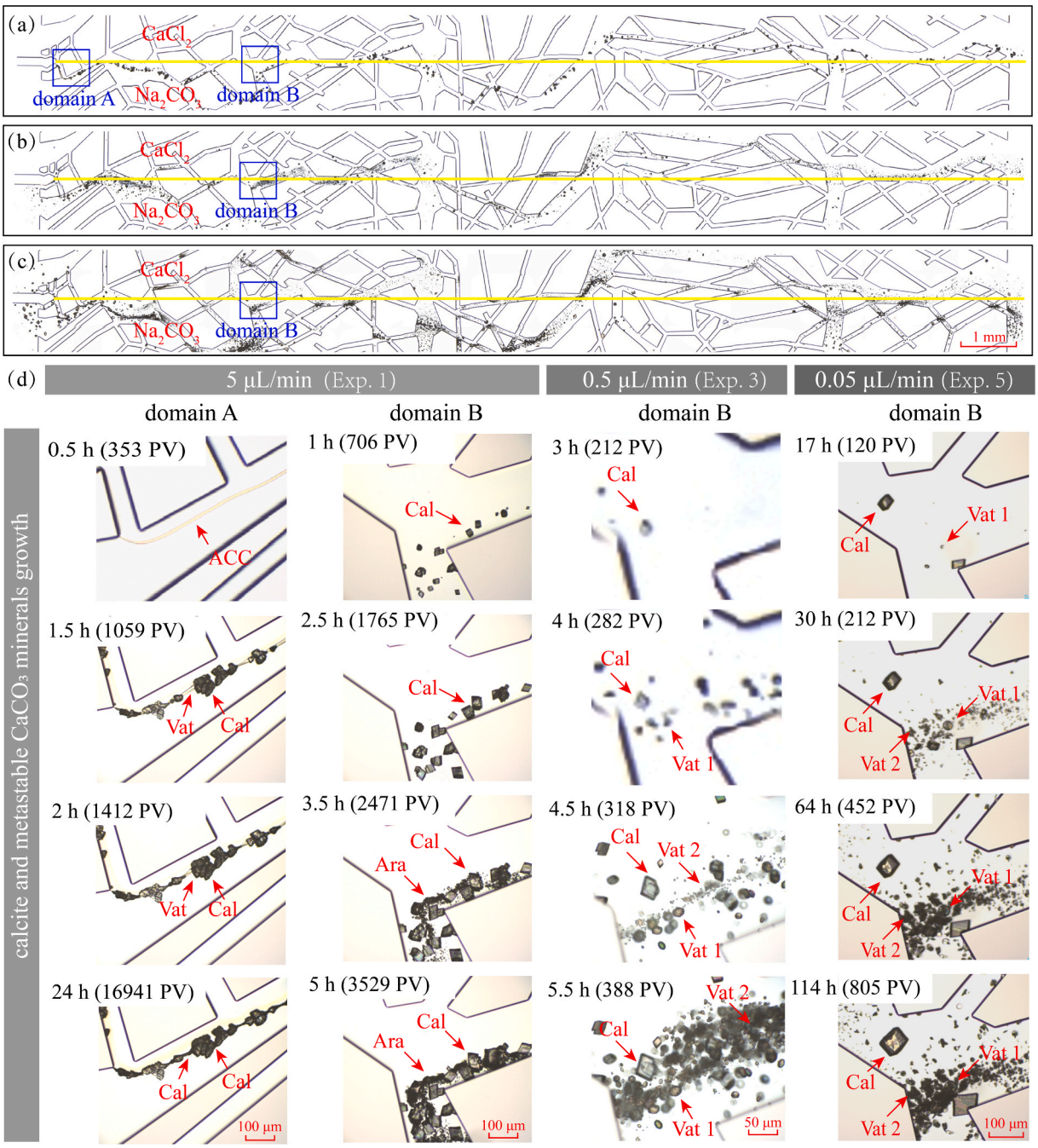

Fig. 5. (a) Locations of microscopic images showing $CaCO_3$ precipitation in plot b. The central line of the micromodel is indicated by yellow lines. (b) Images of $CaCO_3$ minerals in domain A and B at different flow rates over time. Representative images were selected from Exp. 1 for 5 μL/min, Exp. 3 for 0.5 μL/min, and Exp. 5 for 0.05 μL/min. The mineral phases were identified by Raman spectra. ACC: amorphous calcium carbonate, Cal: calcite, Vat: vaterite, and Ara: aragonite.

influenced transport regime. These results are consistent with the D_2O–H_2O mixing experiments, indicating that precipitation-induced flow reorganization directly feeds back on mixing patterns, thereby influencing the spatial distribution of supersaturation and subsequent mineral growth.

Because flow rate is the primary control on mixing geometry and subsequent precipitation patterns, spatial distributions of IAP were evaluated at the initial stage (20 PV) to quantify the geochemical drivers underlying these flow-dependent behaviors (Fig. 8). At comparable injected PV, lower flow rates (0.5 and 0.05 μL/min) resulted in longer reactant residence times and broader mixing zones, producing spatially extensive regions where IAP exceeded the solubility products (K_{sp}) of calcite, vaterite, and ACC. In contrast, at the highest flow rate (5 μL/min), supersaturation was confined to narrower, streamline-aligned zones characterized by sharply elevated IAP values (Fig. 8a).

Although the IAP distributions do not directly mirror the D_2O-H_2O mixing intensity maps, they exhibit strong spatial correspondence in terms of the location, orientation, and flow-rate–dependent width of the dominant mixing bands. Together, the experimental observations and numerical simulations establish a mechanistic link between transport-controlled mixing and subsequent precipitation behavior: broad saturation zones at lower flow rates promote spatially distributed and less predictable precipitation patterns, whereas confined high-IAP zones at higher flow rates favor localized and more reproducible precipitation patterns. This relationship further explains why initial $CaCO_3$ deposition remained comparable across experiments conducted at identical flow

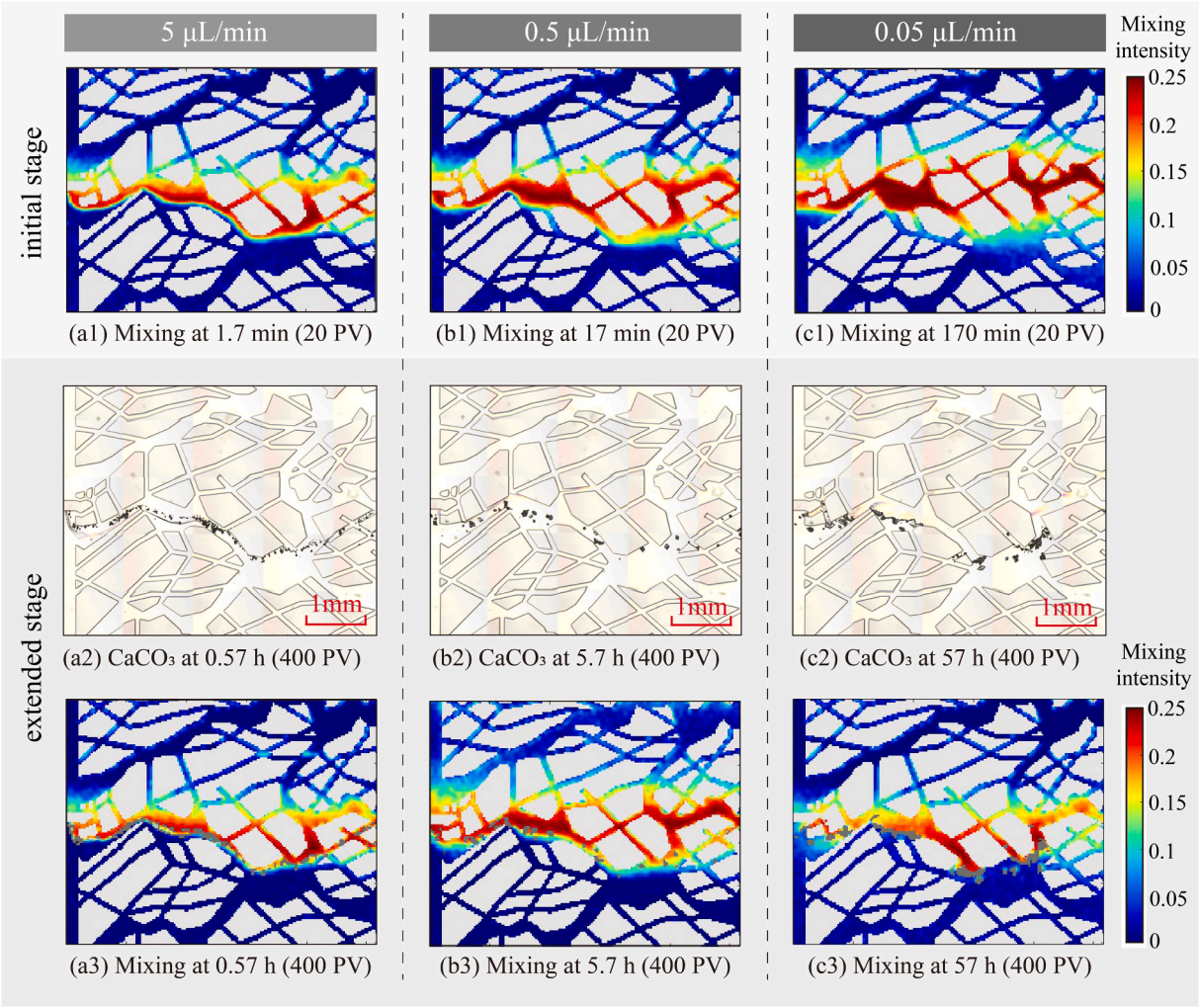

Fig. 6. Influence of $CaCO_3$ precipitation on the evolution of mixing intensity, monitored using D_2O-H_2O tracer experiments, under varying flow rates at early (20 PV) and extended (400 PV) stages. The $CaCO_3$ precipitation in plots b was added to plots c to better demonstrate the impact of precipitation on mixing dynamics over time. The changes in mixing intensity, shown at different time points, highlight how the ongoing $CaCO_3$ precipitation process interacts with and modifies flow behavior at different stages of the experiment.

rates, while pronounced divergence in precipitation amounts emerged at later stages (> 100 PV) as precipitation-induced flow reorganization progressively altered mixing and supersaturation fields (Fig. 3).

The initial-stage IAP distributions further elucidate how flow-controlled mixing governs the sequence and spatial organization of $CaCO_3$ polymorph precipitation. At 20 PV, saturation zones exceeding the K_{sp} of calcite, vaterite, and ACC developed across all three flow rates, indicating that mixing is sufficient to generate thermodynamically favorable conditions for all $CaCO_3$ phases. In most regions, IAP increased progressively with mixing, crossing successive K_{sp} thresholds and favoring the classical precipitation sequence, in which calcite nucleates first, followed by metastable polymorphs (e.g., vaterite, aragonite, and ACC) occupying the remaining pore space. In contrast, precipitation behavior in domain A at higher flow rates (5 and 0.5 μL/min) deviated from this classical sequence. Here, narrow, streamline-shaped saturation zones (~ 13 μm wide) with sharply elevated IAP developed (Fig. 8a and b). Within these zones, local IAP rapidly exceeded the K_{sp} of ACC before calcite nucleation could occur, consistent with the preferential formation of ACC exhibiting streamline-aligned morphologies observed experimentally (Fig. 5b). Although ACC is generally expected to transform rapidly into crystalline polymorphs within minutes under ambient conditions (Shen et al., 2006), it persisted for several hours in our system. This persistence suggests that dynamically generated, flow-controlled high-IAP microenvironments can stabilize ACC by sustaining extreme supersaturation over limited spatial extents, consistent with previous studies showing that spatial confinement delays ACC transformation (Ruzicka, 2008). However, when spatial confinement becomes progressively influenced by ACC precipitation and clogging, mixing is attenuated, and the initially flow-controlled high-IAP microenvironments are disrupted. Under these conditions, ACC becomes thermodynamically unstable and transforms into more stable crystalline $CaCO_3$ polymorphs, such as vaterite and calcite. This ACC-to-crystalline phase transformation is governed by coupled flow–mixing–precipitation processes and thus operates on timescales one to two orders of magnitude longer than molecular-scale mineral phase transformations (Nielsen et al., 2014). Importantly, our results extend this concept by showing that such stabilizing environments can arise transiently from localized flow and mixing, without requiring static geometric confinement.

Together, these results demonstrate that flow-controlled mixing geometry not only determines where supersaturation develops, but also governs which $CaCO_3$ polymorphs nucleate and persist. By coupling hydrodynamic transport with spatially heterogeneous supersaturation fields, flow rate exerts a critical control on both the reproducibility and variability of $CaCO_3$ precipitation pathways in fractured media.

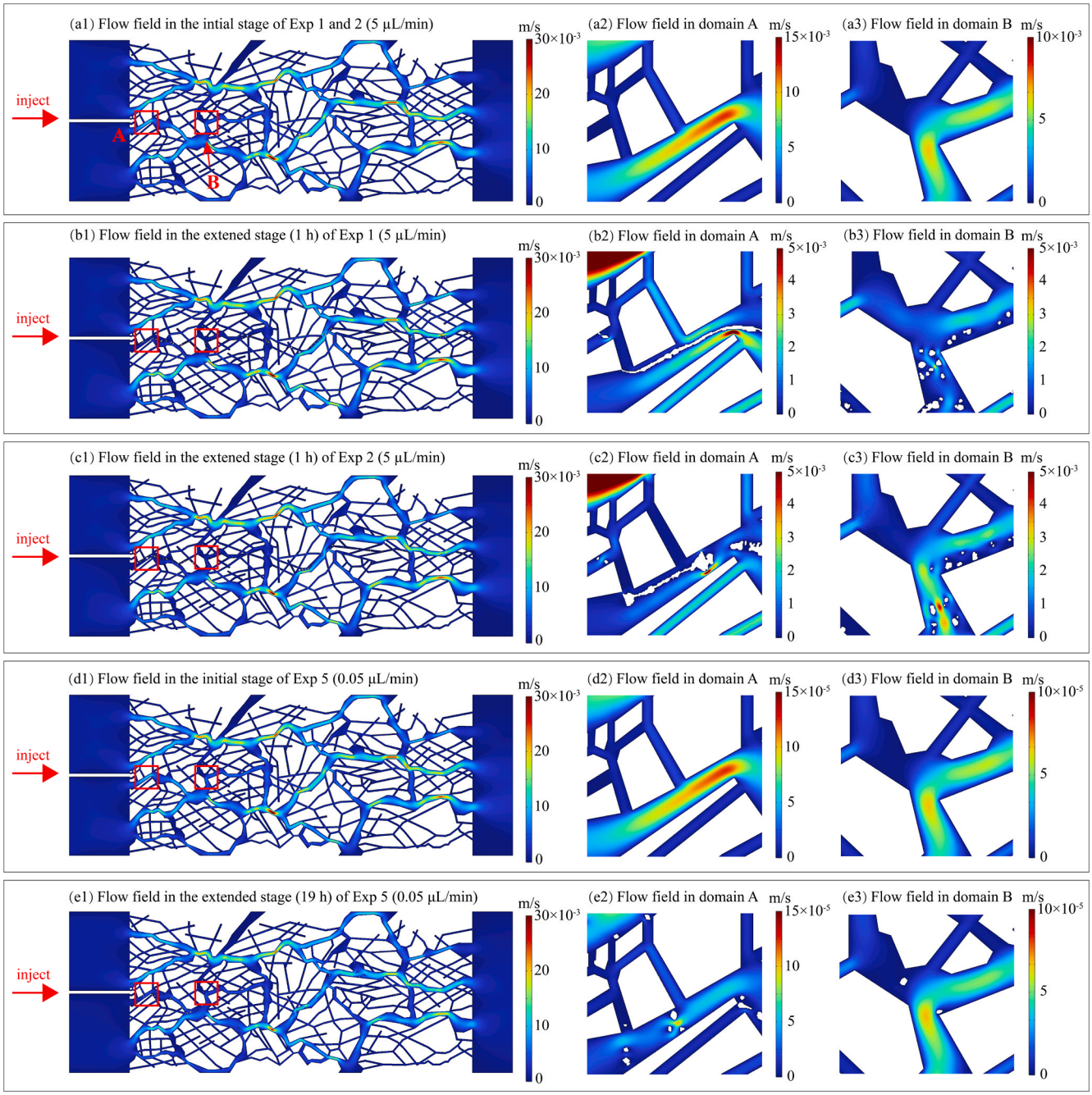

Fig. 7. Steady-state flow fields in the heterogeneous fracture at the onset of the initial and extended stages. (a-c) Exp. 1 and 2 at 5 μL/min: (a) onset of the initial stage; (b, c) onset of the extended stage. (d, e) Exp. 5 at 0.05 μL/min: (d) onset of the initial stage; (e) onset of the extended stage. The zoomed-in images on the right (columns 2 and 3) show detailed flow fields in domain A and domain B, providing a closer view of the flow dynamics in these regions at different stages of the experiment.

3.4. *Influence of flow-controlled mixing on nucleation*

Building on previous findings that highlight the role of flow-driven mixing in the location and persistence of supersaturation, we now investigate how flow-controlled mixing dynamics govern nucleation in $CaCO_3$ precipitation. Nucleation is inherently stochastic, leading to variability even under identical macroscopic conditions. This randomness significantly influences early nucleation, which in turn affects subsequent crystal growth, structural evolution, and overall precipitation. To explore the early-stage nucleation process, Darcy-scale advection–diffusion-nucleation simulations were conducted across varying flow rates.

Due to the limitations of Darcy-scale simulations, it is challenging to explicitly resolve the detailed saturation conditions favoring metastable phases like ACC. Furthermore, although streamline-shaped ACC was observed in certain high-flow locations, experimental observations indicate that calcite is the dominant phase in most regions during early nucleation. Therefore, focusing on calcite allows us to simplify the analysis without sacrificing relevance, enabling the isolation of key transport-driven processes governing precipitation onset. For each flow rate, ten simulations were performed over a duration of 1 h. The $CaCO_3$ content in each grid cell, normalized by the total $CaCO_3$ content (Nu/Nu_T), was used as a metric for nucleation. Representative results are shown in Fig. 9, with the left panels show the nucleation distribution for

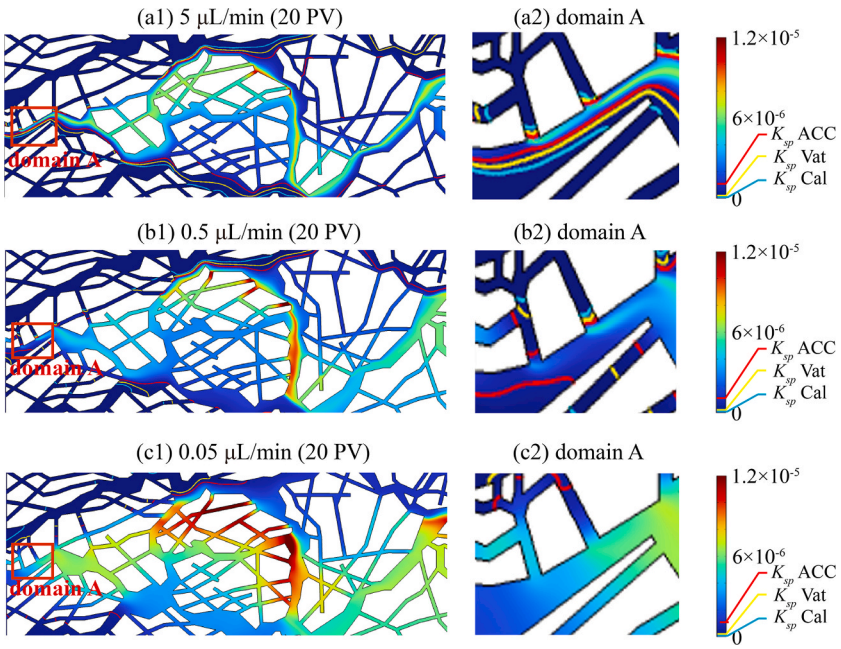

Fig. 8. Simulated ion activity products (IAP) of $CaCO_3$ under different flow rates at the initial stage prior to $CaCO_3$ precipitation. The IAP values exceeding the solubility product (K_{sp}) of calcite ($10^{-8.48}$), vaterite ($10^{-7.14}$), and ACC ($10^{-6.5}$) indicate thermodynamic oversaturation (Plummer and Busenberg, 1982), suggesting the potential for mineral precipitation. Distinct patterns of oversaturation are observed at different flow rates, reflecting the impact of flow dynamics on ion concentration and mineral formation. The zoomed-in images on the right (domain A) provide a detailed view of the oversaturation patterns.

one representative case, while the right panels display the integrated nucleation along the x-direction for each y-position, based on five representative cases.

As shown in Fig. 9 (panels a1-c2), nucleation locations are primarily constrained by the flow-controlled saturation zone. At a high flow rate of 5 μL/min (panel a1), nucleation occurs rapidly and is concentrated within a narrow region, where chemical gradients are steeper. In contrast, at lower flow rates (0.5 μL/min and 0.05 μL/min), nucleation is significantly reduced and more dispersed within a broader region. In a complete precipitation process involving both nucleation and subsequent crystal growth, the formation of the initial nucleation zone may give rise to a localized "shadow zone," where crystal growth depletes supersaturation and suppresses further nucleation (Steefel and Yang, 2021). The subsequent spatial distribution of crystals thus reflects the structure of the initial nucleation zone. Although crystal growth was not explicitly included in the simulations, the experimentally observed dispersion of precipitates during the extended stage is primarily governed by the probabilistic nucleation pattern established during the initial stage. This behavior is consistent with the wider saturation zones and slower accumulation of supersaturation observed under these conditions. The integration of nucleation along the y-axis (panels a2-c2) further illustrates the above phenomenon: as the flow rate decreases, nucleation activity spreads across a larger portion of the domain.

Given the inherent randomness of nucleation, there is noticeable variability in nucleation amounts and distributions across different simulation cases (Fig. 9 a2-c2). The integration of nucleation rate along the y-axis reveals distinct patterns for each simulation case, with all parameters held constant except for the random distribution of induction times (τ_p) relative to the CNT theoretical induction time (τ_N). Within the same simulation case, nucleation amounts along the y-direction also exhibit randomness, with the highest nucleation not necessarily occurring at the center of the mixing zone, where the saturation index would be the highest. This randomness in nucleation highlights that it can have a more significant impact than the flow-controlled saturation conditions, emphasizing the importance of stochastic nature of the nucleation process.

These results demonstrate that flow-controlled mixing governs the spatial extent and distribution of nucleation within fractured media. Higher flow rates lead to more localized nucleation, while lower flow rates result in higher variability and a broader nucleation distribution (Fig. 9 c2), suggesting that random fluctuations have a greater influence on nucleation in slower-flow environments. This variability is crucial for understanding how initial nucleation patterns impact the growth of $CaCO_3$ crystals and the final precipitation amounts, as flow-driven mixing not only regulates supersaturation development but also controls the formation and persistence of nucleation sites, shaping $CaCO_3$ precipitation pathways.

4. Conclusion

This study provides important insights into $CaCO_3$ precipitation within heterogeneous fracture networks, with broad implications for aquifers, geothermal reservoirs, and CO_2 storage sites. We show that the interplay between flow transport, chemical mixing, nucleation variability, and mineral growth dynamics governs precipitation patterns, which in turn reshape fracture geometry, permeability, and total mineral accumulation. At high flow rates, rapid ion replenishment confines precipitation to narrow zones and yields more reproducible patterns, but accelerated calcite growth can induce early clogging that reduces storage capacity. In contrast, low flow rates, solute transport is dominated by transverse diffusion, producing broader saturation zones where precipitation is slower, more variable, and strongly influenced by stochastic nucleation. This heterogeneity generates uneven permeability, with some regions clogging while others remain open, directly affecting long-term fluid flow and solute transport.

Early-stage processes—such as stochastic nucleation, transient ACC stabilization, and polymorph transitions—may not change the final mineralogy but strongly shape fracture evolution and local saturation index distributions. These dynamics are crucial for understanding mineral trapping efficiency and the spatial distribution of stored CO_2, making them critical for optimizing geological storage. Recognizing the influence of flow rate and transport on precipitation can guide injection strategies to minimize clogging and improve storage ability.

These findings offer valuable insights into fracture sealing and

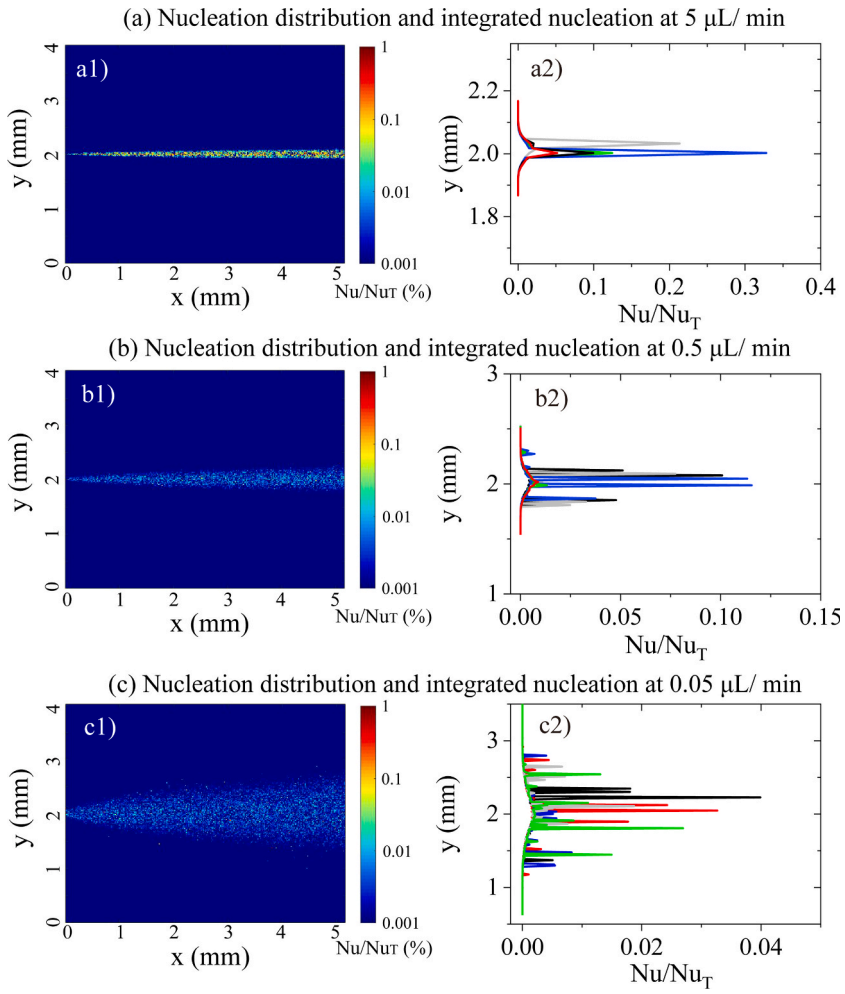

Fig. 9. Spatial nucleation distribution (a1-c1) and integrated nucleation across the x-axis (a2-c2) at varying flow rates: (a) 5 μL/min, (b) 0.5 μL/min, and (c) 0.05 μL/min. In the right panels, different colors represent the nucleation results for five distinct cases.

mineral trapping in CO_2 storage and hydrothermal fracture evolution. In both of these geological processes, fracture sealing plays a critical role in controlling the efficiency of CO_2 storage and fluid migration in hydrothermal systems. We demonstrate that flow rate, stochastic nucleation, and mineral polymorphs govern the rate and predictability of fracture sealing. High flow rates tend to cause rapid fracture blockage, restricting further mineral growth and fluid-rock interactions. In contrast, slower, heterogeneous precipitation under low flow conditions helps preserve flow pathways, enabling continued fluid-rock interactions. However, low flow rates introduce greater uncertainty in sealing efficiency. Dense calcite formation can obstruct flow, reducing mineral trapping capacity, while metastable phases such as vaterite, aragonite, and ACC form more porous structures, which facilitate continued ion transport and precipitation. By controlling injection concentration, we can favor the formation of metastable phases, promoting mineral diversity. These dynamics are crucial for understanding the evolution of fracture networks in CO_2 storage and hydrothermal systems, where mineral precipitation governs fluid flow and fracture permeability over time.

From a modeling perspective, our findings explain the challenges in capturing precipitation dynamics in heterogeneous fractures: complex geometries lead to heterogeneous local saturation distributions, stochastic nucleation, polymorph effects, and strong flow-precipitation feedbacks. Future models should therefore incorporate fracture-scale heterogeneity, random nucleation processes, multi-phase $CaCO_3$ dynamics, and coupled flow–reaction–geometry interactions to realistically predict precipitation patterns and their long-term impacts on permeability and storage efficiency. These insights enhance our understanding of mineral precipitation in fractured networks and provide valuable context for future studies on flow-dependent precipitation processes.

Data availability

Data are available through Mendeley Data at: https://doi.org/10.17632/xhd3z47wc5.1.

CRediT authorship contribution statement

Tengfei Ma: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Hang Deng:** Methodology. **Puxuan Kang:** Formal analysis. **Yuchao Feng:** Data curation. **Xiaofan Yang:** Methodology, Conceptualization. **Liang Zhao:** Writing – review & editing. **Xueying Li:** Methodology. **Dexi Wang:** Funding acquisition. **Xingming Chen:** Funding acquisition. **Zhengmao Yang:** Funding acquisition. **Weisheng Qiu:** Project administration. **Junfeng Ji:** Writing – review & editing. **Yuanyuan Liu:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research is funded by the National Natural Science Foundation of China (Grant No. 42273054 and 42373055), the Special fund for Carbon Peak and Carbon Neutrality Science and Technology Innovation of Jiangsu Province in 2022 (BE2022603) and 2025 (BT2025016), and Jiangsu 'Double Innovation Plan'.

References

Ahkami, M., Parmigiani, A., Di Palma, P.R., Saar, M.O., Kong, X.-Z., 2020. A lattice-Boltzmann study of permeability-porosity relationships and mineral precipitation patterns in fractured porous media. Comput. Geosci. 24, 1865–1882.

Ahoulou, A.W.A., Tinet, A.-J., Oltéan, C., Golfier, F., 2020. Experimental insights into the interplay between buoyancy, convection, and dissolution reaction. J. Geophys. Res. Solid Earth 125, e2020JB020854.

Algive, L., Bekri, S., Vizika, O., 2009. Reactive Pore Network Modeling Dedicated to the Determination of the Petrophysical Property Changes while Injecting CO_2, in: SPE Annual Technical Conference and Exhibition. Presented at the SPE Annual Technical Conference and Exhibition, SPE, New Orleans, Louisiana, p. SPE-124305-MS. DOI: 10.2118/124305-MS.

Anders, M.H., Laubach, S.E., Scholz, C.H., 2014. Microfractures: a review. J. Struct. Geol. 69, 377–394.

Avaro, J.T., Wolf, S.L.P., Hauser, K., Gebauer, D., 2020. Stable prenucleation calcium carbonate clusters define liquid–liquid phase separation. Angew. Chem. Int. Ed. 59, 6155–6159.

Bashir, A., Ali, M., Patil, S., Aljawad, M.S., Mahmoud, M., Al-Shehri, D., Hoteit, H., Kamal, M.S., 2024. Comprehensive review of CO_2 geological storage: exploring principles, mechanisms, and prospects. Earth Sci. Rev. 249, 104672.

Boyd, V., Yoon, H., Zhang, C., Oostrom, M., Hess, N., Fouke, B., Valocchi, A.J., Werth, C. J., 2014. Influence of Mg^{2+} on $CaCO_3$ precipitation during subsurface reactive transport in a homogeneous silicon-etched pore network. Geochim. Cosmochim. Acta 135, 321–335.

Caupin, F., Grisenti, R.E., 2026. Nucleation statistics from experiments as a benchmark for theory and simulations. J. Phys. Chem. Lett. 17, 1877–1891.

Chomsurin, C., Werth, C.J., 2003. Analysis of pore-scale nonaqueous phase liquid dissolution in etched silicon pore networks. Water Resour. Res. 39, 2002WR001643.

De Silva, G.P.D., Ranjith, P.G., Perera, M.S.A., 2015. Geochemical aspects of CO_2 sequestration in deep saline aquifers: a review. Fuel 155, 128–143.

De Yoreo, J.J., 2003. Principles of crystal nucleation and growth. Rev. Mineral. Geochem. 54, 57–93.

Deng, H., Poonoosamy, J., Molins, S., 2022. A reactive transport modeling perspective on the dynamics of interface-coupled dissolution-precipitation. Appl. Geochem. 137, 105207.

Deng, H., Steefel, C., Molins, S., DePaolo, D., 2018. Fracture evolution in multimineral systems: the role of mineral composition, flow rate, and fracture aperture heterogeneity. ACS Earth Space Chem. 2, 112–124.

Deng, H., Tournassat, C., Molins, S., Claret, F., Steefel, C.I., 2021. A pore-scale investigation of mineral precipitation driven diffusivity change at the column-scale. Water Resour. Res. 57, e2020WR028483.

Fazeli, H., Masoudi, M., Patel, R.A., Aagaard, P., Hellevang, H., 2020. Pore-scale modeling of nucleation and growth in porous media. ACS Earth Space Chem. 4, 249–260.

Fernandez-Martinez, A., Hu, Y., Lee, B., Jun, Y.-S., Waychunas, G.A., 2013. In situ determination of interfacial energies between heterogeneously nucleated $CaCO_3$ and quartz substrates: thermodynamics of CO_2 mineral trapping. Environ. Sci. Technol. 47, 102–109.

Gebauer, D., Völkel, A., Cölfen, H., 2008. Stable prenucleation calcium carbonate clusters. Science 322, 1819–1822.

Geiger, S., Cortis, A., Birkholzer, J.T., 2010. Upscaling solute transport in naturally fractured porous media with the continuous time random walk method. Water Resour. Res. 46, 2010WR009133.

Gislason, S.R., Oelkers, E.H., 2014. Carbon storage in basalt. Science 344, 373–374.

Godoy, P.A., Orujov, A., Pérez Gramatges, A., Aryana, S.A., 2025. Rock-on-a-chip: a novel method for designing representative microfluidic platforms based on real rock structures and pore network modelling. Lab Chip 25, 3109–3122.

Goh, L., Chen, K., Bhamidi, V., He, G., Kee, N.C.S., Kenis, P.J.A., Zukoski, C.F., Braatz, R. D., 2010. A stochastic model for nucleation kinetics determination in droplet-based microfluidic systems. Cryst. Growth Des. 10, 2515–2521.

Gupta, N., Verma, A., 2023. Supercritical carbon dioxide utilization for hydraulic fracturing of shale reservoir, and geo-storage: a review. Energy Fuels 37, 14604–14621.

Han, Y.S., Hadiko, G., Fuji, M., Takahashi, M., 2005. Effect of flow rate and CO_2 content on the phase and morphology of $CaCO_3$ prepared by bubbling method. J. Cryst. Growth 276, 541–548.

Haynes, W.M. (Ed.), 2014. CRC Handbook of Chemistry and Physics, 0 ed. CRC Press.

Henzler, K., Fetisov, E.O., Galib, M., Baer, M.D., Legg, B.A., Borca, C., Xto, J.M., Pin, S., Fulton, J.L., Schenter, G.K., Govind, N., Siepmann, J.I., Mundy, C.J., Huthwelker, T., De Yoreo, J.J., 2018. Supersaturated calcium carbonate solutions are classical. Sci. Adv. 4, eaao6283.

Jiang, Q.-R., Hu, R., Deng, H., Ling, B., Yang, Z., Chen, Y.-F., 2025. Controls of the nucleation rate and advection rate on barite precipitation in fractured porous media. Langmuir 41, 1250–1259.

Jin, B., Chen, Y., Pyles, H., Baer, M.D., Legg, B.A., Wang, Z., Washton, N.M., Mueller, K. T., Baker, D., Schenter, G.K., Mundy, C.J., De Yoreo, J.J., 2025. Formation, chemical evolution and solidification of the dense liquid phase of calcium (bi)carbonate. Nat. Mater. 24, 125–132.

Jun, Y.-S., Zhu, Y., Wang, Y., Ghim, D., Wu, X., Kim, D., Jung, H., 2022. Classical and nonclassical nucleation and growth mechanisms for nanoparticle formation. Annu. Rev. Phys. Chem. 73, 453–477.

Lassin, A., André, L., Devau, N., Lach, A., Beuvier, T., Gibaud, A., Gaboreau, S., Azaroual, M., 2018. Dynamics of calcium carbonate formation: geochemical modeling of a two-step mechanism. Geochim. Cosmochim. Acta 240, 236–254.

Laubach, S.E., Lander, R.H., Criscenti, L.J., Anovitz, L.M., Urai, J.L., Pollyea, R.M., Hooker, J.N., Narr, W., Evans, M.A., Kerisit, S.N., Olson, J.E., Dewers, T., Fisher, D., Bodnar, R., Evans, B., Dove, P., Bonnell, L.M., Marder, M.P., Pyrak-Nolte, L., 2019. The role of chemistry in fracture pattern development and opportunities to advance interpretations of geological materials. Rev. Geophys. 57, 1065–1111.

Li, L., Steefel, C.I., Yang, L., 2008. Scale dependence of mineral dissolution rates within single pores and fractures. Geochim. Cosmochim. Acta 72, 360–377.

Li, Q., Fernandez-Martinez, A., Lee, B., Waychunas, G.A., Jun, Y.-S., 2014. Interfacial energies for heterogeneous nucleation of calcium carbonate on mica and quartz. Environ. Sci. Technol. 48, 5745–5753.

Li, Q., Jun, Y.-S., 2018. The apparent activation energy and pre-exponential kinetic factor for heterogeneous calcium carbonate nucleation on quartz. Commun. Chem. 1, 56.

Li, Q., Steefel, C.I., Jun, Y.-S., 2017. Incorporating nanoscale effects into a continuum-scale reactive transport model for CO_2 -deteriorated cement. Environ. Sci. Technol. 51, 10861–10871.

Liu, Y., Zhang, C., Hilpert, M., Kuhlenschmidt, M.S., Kuhlenschmidt, T.B., Nguyen, T.H., 2012. Transport of *Cryptosporidium parvum* oocysts in a silicon micromodel. Environ. Sci. Technol. 46, 1471–1479.

Lönartz, M.I., Yang, Y., Deissmann, G., Bosbach, D., Poonoosamy, J., 2023. Capturing the dynamic processes of porosity clogging. Water Resour. Res. 59, e2023WR034722.

Lu, P., Apps, J., Zhang, G., Gysi, A., Zhu, C., 2024. Knowledge gaps and research needs for modeling CO_2 mineralization in the basalt-CO_2-water system: a review of laboratory experiments. Earth Sci. Rev. 254, 104813.

Lyu, Q., Tan, J., Li, L., Ju, Y., Busch, A., Wood, D.A., Ranjith, P.G., Middleton, R., Shu, B., Hu, C., Wang, Z., Hu, R., 2021. The role of supercritical carbon dioxide for recovery of shale gas and sequestration in gas shale reservoirs. Energy Environ. Sci. 14, 4203–4227.

Matter, J.M., Stute, M., Snæbjörnsdottir, S.Ó., Oelkers, E.H., Gislason, S.R., Aradottir, E. S., Sigfusson, B., Gunnarsson, I., Sigurdardottir, H., Gunnlaugsson, E., Axelsson, G., Alfredsson, H.A., Wolff-Boenisch, D., Mesfin, K., Taya, D.F.D.L.R., Hall, J., Dideriksen, K., Broecker, W.S., 2016. Rapid carbon mineralization for permanent disposal of anthropogenic carbon dioxide emissions. Science 352, 1312–1314.

Mi, L., Fan, H., Fan, T., Gong, L., Niu, T., Su, X., Luo, J., Sun, Y., 2023. Development characteristics of multi-scale fracture network systems in metamorphic buried hills. Front. Earth Sci. 10, 1108032.

Molins, S., 2015. Reactive interfaces in direct numerical simulation of pore-scale processes. Rev. Mineral. Geochem. 80, 461–481.

Molins, S., Trebotich, D., Yang, L., Ajo-Franklin, J.B., Ligocki, T.J., Shen, C., Steefel, C.I., 2014. Pore-scale controls on calcite dissolution rates from flow-through laboratory and numerical experiments. Environ. Sci. Technol. 48, 7453–7460.

Morais, S., Cario, A., Liu, N., Bernard, D., Lecoutre, C., Garrabos, Y., Ranchou-Peyruse, A., Dupraz, S., Azaroual, M., Hartman, R.L., Marre, S., 2020. Studying key processes related to CO_2 underground storage at the pore scale using high pressure micromodels. React. Chem. Eng. 5, 1156–1185.

Morris, P.D., McPherson, I.J., Meloni, G.N., Unwin, P.R., 2020. Nanoscale kinetics of amorphous calcium carbonate precipitation in H_2O and D_2O. PCCP 22, 22107–22115.

Morse, J.W., Arvidson, R.S., Lüttge, A., 2007. Calcium carbonate formation and dissolution. Chem. Rev. 107, 342–381.

Mostaghimi, P., Liu, M., Arns, C.H., 2016. Numerical simulation of reactive transport on Micro-CT images. Math. Geosci. 48, 963–983.

Neuman, S.P., 2005. Trends, prospects and challenges in quantifying flow and transport through fractured rocks. Hydrgeol. J. 13, 124–147.

Nguyen, T.T.T., Shoukry, A.E., Saraji, S., 2025. Investigating reactive transport and precipitation patterns of calcium carbonate in fractured porous media. J. Colloid Interface Sci. 679, 467–480.

Nielsen, M.H., Aloni, S., De Yoreo, J.J., 2014. In situ TEM imaging of $CaCO_3$ nucleation reveals coexistence of direct and indirect pathways. Science 345, 1158–1162.

Nisbet, H., Buscarnera, G., Carey, J.W., Chen, M.A., Detournay, E., Huang, H., Hyman, J. D., Kang, P.K., Kang, Q., Labuz, J.F., Li, W., Matter, J., Neil, C.W., Srinivasan, G., Sweeney, M.R., Voller, V.R., Yang, W., Yang, Y., Viswanathan, H.S., 2024. Carbon mineralization in fractured mafic and ultramafic rocks: a review. Rev. Geophys. 62, e2023RG000815.

Nisbet, H., Li, R., Purswani, P., Chen, M., Yang, W., Neil, C., Kang, Q., Mohanty, K., Kang, P.K., Viswanathan, H., 2025. Mechanistic understanding of carbon mineralization in fracture systems using microfluidics. Lab Chip 25, 4024–4037.

Noiriel, C., Gouze, P., Madé, B., 2013. 3D analysis of geometry and flow changes in a limestone fracture during dissolution. J. Hydrol. 486, 211–223.

Noiriel, C., Steefel, C.I., Yang, L., Ajo-Franklin, J., 2012. Upscaling calcium carbonate precipitation rates from pore to continuum scale. Chem. Geol. 318–319, 60–74.

Noiriel, C., Steefel, C.I., Yang, L., Bernard, D., 2016. Effects of pore-scale precipitation on permeability and flow. Adv. Water Resour. 95, 125–137.
Nooraiepour, M., Masoudi, M., Hellevang, H., 2021a. Probabilistic nucleation governs time, amount, and location of mineral precipitation and geometry evolution in the porous medium. Sci. Rep. 11, 16397.
Nooraiepour, M., Masoudi, M., Shokri, N., Hellevang, H., 2021b. Probabilistic nucleation and crystal growth in porous medium: new insights from calcium carbonate precipitation on primary and secondary substrates. ACS Omega 6, 28072–28083.
Olson, J.E., Laubach, S.E., Lander, R.H., 2009. Natural fracture characterization in tight gas sandstones: integrating mechanics and diagenesis. AAPG Bull. 93, 1535–1549.
Plummer, L.N., Busenberg, E., 1982. The solubilities of calcite, aragonite and vaterite in CO_2-H_2O solutions between 0 and 90 °C, and an evaluation of the aqueous model for the system $CaCO_3$-CO_2-H_2O. Geochim. Cosmochim. Acta 46, 1011–1040.
Poonoosamy, J., Westerwalbesloh, C., Deissmann, G., Mahrous, M., Curti, E., Churakov, S.V., Klinkenberg, M., Kohlheyer, D., Von Lieres, E., Bosbach, D., Prasianakis, N.I., 2019. A microfluidic experiment and pore scale modelling diagnostics for assessing mineral precipitation and dissolution in confined spaces. Chem. Geol. 528, 119264.
Raza, A., Glatz, G., Gholami, R., Mahmoud, M., Alafnan, S., 2022. Carbon mineralization and geological storage of CO_2 in basalt: mechanisms and technical challenges. Earth Sci. Rev. 229, 104036.
Roman, S., Soulaine, C., AlSaud, M.A., Kovscek, A., Tchelepi, H., 2016. Particle velocimetry analysis of immiscible two-phase flow in micromodels. Adv. Water Resour. 95, 199–211.
Ruzicka, M.C., 2008. On dimensionless numbers. Chem. Eng. Res. Des. 86, 835–868.
San, X., Hu, J., Chen, M., Niu, H., Smeets, P.J.M., Malliakas, C.D., Deng, J., Koo, K., Dos Reis, R., Dravid, V.P., Hu, X., 2023. Unlocking the mysterious polytypic features within vaterite CaCO3. Nat. Commun. 14, 7858.
Sanglas, J., Trinchero, P., Painter, S.L., Cvetkovic, V., Poteri, A., Selroos, J., Zou, L., 2024. Significance of low-velocity zones on solute retention in rough fractures. Water Resour. Res. 60, e2023WR036221.
Schuszter, G., Brau, F., De Wit, A., 2016. Flow-driven control of calcium carbonate precipitation patterns in a confined geometry. PCCP 18, 25592–25600.
Shapiro, A.M., 2001. Effective matrix diffusion in kilometer-scale transport in fractured crystalline rock. Water Resour. Res. 37, 507–522.
Shapiro, A.M., Hsieh, P.A., Burton, W.C., Walsh, G.J., 2007. Integrated multi-scale characterization of ground-water flow and chemical transport in fractured crystalline rock at the Mirror Lake Site, New Hampshire, in: Hyndman, D.W., Day-Lewis, F.D., Singha, K. (Eds.), Geophysical Monograph Series. American Geophysical Union, Washington, D. C., pp. 201–225.
Shen, Q., Wei, H., Zhou, Y., Huang, Y., Yang, H., Wang, D., Xu, D., 2006. Properties of amorphous calcium carbonate and the template action of vaterite spheres. J. Phys. Chem. B 110, 2994–3000.
Song, K., Bang, J.-H., Chae, S.-C., Kim, J., Lee, S.-W., 2022. Phase and morphology of calcium carbonate precipitated by rapid mixing in the absence of additives. RSC Adv. 12, 19340–19349.
Song, W., De Haas, T.W., Fadaei, H., Sinton, D., 2014. Chip-off-the-old-rock: the study of reservoir-relevant geological processes with real-rock micromodels. Lab Chip 14, 4382–4390.
Soulaine, C., 2024. Micro-continuum modeling: an hybrid-scale approach for solving coupled processes in porous media. Water Resour. Res. 60, e2023WR035908.
Soulaine, C., Maes, J., Roman, S., 2021. Computational microfluidics for geosciences. Front. Water 3, 643714.
Soulaine, C., Roman, S., Kovscek, A., Tchelepi, H.A., 2017. Mineral dissolution and wormholing from a pore-scale perspective. J. Fluid Mech. 827, 457–483.
Starchenko, V., Ladd, A.J.C., 2018. The development of wormholes in laboratory-scale fractures: perspectives from three-dimensional simulations. Water Resour. Res. 54, 7946–7959.
Steefel, C., Depaolo, D., Lichtner, P., 2005. Reactive transport modeling: an essential tool and a new research approach for the Earth sciences. Earth Planet. Sci. Lett. 240, 539–558.
Steefel, C.I., Appelo, C.A.J., Arora, B., Jacques, D., Kalbacher, T., Kolditz, O., Lagneau, V., Lichtner, P.C., Mayer, K.U., Meeussen, J.C.L., Molins, S., Moulton, D., Shao, H., Šimůnek, J., Spycher, N., Yabusaki, S.B., Yeh, G.T., 2015. Reactive transport codes for subsurface environmental simulation. Comput. Geosci. 19, 445–478.
Steefel, C.I., Hu, M., 2022. Reactive transport modeling of mineral precipitation and carbon trapping in discrete fracture networks. Water Resour. Res. 58, e2022WR032321.
Steefel, C.I., Lasaga, A.C., 1994. A coupled model for transport of multiple chemical species and kinetic precipitation/dissolution reactions with application to reactive flow in single phase hydrothermal systems. Am. J. Sci. 294, 529–592.
Steefel, C.I., Lichtner, P.C., 1994. Diffusion and reaction in rock matrix bordering a hyperalkaline fluid-filled fracture. Geochim. Cosmochim. Acta 58, 3595–3612.
Steefel, C.I., Van Cappellen, P., 1990. A new kinetic approach to modeling water-rock interaction: the role of nucleation, precursors, and Ostwald ripening. Geochim. Cosmochim. Acta 54, 2657–2677.
Steefel, C.I., Yang, L., 2021. Secondary magnesite formation from forsterite under CO2 sequestration conditions via coupled heterogeneous nucleation and crystal growth. Geochim. Cosmochim. Acta 311, 29–42.
Sulpis, O., Agrawal, P., Wolthers, M., Munhoven, G., Walker, M., Middelburg, J.J., 2022. Aragonite dissolution protects calcite at the seafloor. Nat. Commun. 13, 1104.
Tan, J., Tartakovsky, A.M., Ferris, K., Ryan, E.M., 2016. Investigating the effects of anisotropic mass transport on dendrite growth in high energy density lithium batteries. J. Electrochem. Soc. 163, A318–A327.
Tournassat, C., Steefel, C.I., 2019. Reactive transport modeling of coupled processes in nanoporous media. Rev. Mineral. Geochem. 85, 75–109.
Viswanathan, H.S., Ajo-Franklin, J., Birkholzer, J.T., Carey, J.W., Guglielmi, Y., Hyman, J.D., Karra, S., Pyrak-Nolte, L.J., Rajaram, H., Srinivasan, G., Tartakovsky, D.M., 2022. From fluid flow to coupled processes in fractured rock: recent advances and new frontiers. Rev. Geophys. 60, e2021RG000744.
Voorhees, P.W., 1985. The theory of Ostwald ripening. J. Stat. Phys. 38, 231–252.
Waber, H.N., Smellie, J.A.T., 2008. Characterisation of pore water in crystalline rocks. Appl. Geochem. 23, 1834–1861.
Wang, J., Yu, Q., 2018. Experimental investigations of the process of carbonate fracture dissolution enlargement under reservoir temperature and pressure conditions. Geofluids 2018, 1–19.
Wang, Z., Hu, M., Steefel, C., 2024. Pore-scale modeling of reactive transport with coupled mineral dissolution and precipitation. Water Resour. Res. 60, e2023WR036122.
Weinhardt, F., Class, H., Vahid Dastjerdi, S., Karadimitriou, N., Lee, D., Steeb, H., 2021. Experimental methods and imaging for enzymatically induced calcite precipitation in a microfluidic cell. Water Resour. Res. 57, e2020WR029361.
Weinhardt, F., Deng, J., Hommel, J., Vahid Dastjerdi, S., Gerlach, R., Steeb, H., Class, H., 2022. Spatiotemporal distribution of precipitates and mineral phase transition during biomineralization affect porosity–permeability relationships: microfluidic investigations. Transp. Porous Media 143, 527–549.
Wetzel, M., Kempka, T., Kühn, M., 2022. Hysteresis in permeability evolution simulated for a sandstone by mineral precipitation and dissolution. Adv. Geosci. 58, 1–10.
White, S.K., Spane, F.A., Schaef, H.T., Miller, Q.R.S., White, M.D., Horner, J.A., McGrail, B.P., 2020. Quantification of CO_2 mineralization at the wallula basalt pilot project. Environ. Sci. Technol. 54, 14609–14616.
Xiao, Y., Zhao, C., Cui, H., Chen, Y., Wu, B., Liu, H., 2025. Microscale insights into enzyme-induced carbonate precipitation in rock-based microfluidic chips. Géotechnique 75, 846–857.
Xu, J., Balhoff, M.T., 2023. Emergence of power-law particle size distribution in microfluidic calcium carbonate precipitation: an extended yule process with a ripening effect. Phys. Rev. Lett. 131, 034001.
Xu, R., Li, R., Ma, J., He, D., Jiang, P., 2017. Effect of mineral dissolution/precipitation and CO_2 exsolution on CO_2 transport in geological carbon storage. Acc. Chem. Res. 50, 2056–2066.
Yang, F., Yuan, K., Stack, A.G., Starchenko, V., 2022. Numerical study of mineral nucleation and growth on a substrate. ACS Earth Space Chem. 6, 1655–1665.
Yang, W., Chen, M.A., Lee, S.H., Kang, P.K., 2024. Fluid inertia controls mineral precipitation and clogging in pore to network-scale flows. Proc. Natl. Acad. Sci. 121, e2401318121.
Yoon, H., Chojnicki, K.N., Martinez, M.J., 2019. Pore-scale analysis of calcium carbonate precipitation and dissolution kinetics in a microfluidic device. Environ. Sci. Technol. 53, 14233–14242.
Yoon, H., Valocchi, A.J., Werth, C.J., Dewers, T., 2012. Pore-scale simulation of mixing-induced calcium carbonate precipitation and dissolution in a microfluidic pore network. Water Resour. Res. 48, 2011WR011192.
Zeng, L., Sarmadivaleh, M., Saeedi, A., Chen, Y., Zhong, Z., Xie, Q., 2023. Storage integrity during underground hydrogen storage in depleted gas reservoirs. Earth Sci. Rev. 247, 104625.
Zeng, Y., Cao, J., Wang, Z., Guo, J., Lu, J., 2018. Formation of amorphous calcium carbonate and its transformation mechanism to crystalline $CaCO_3$ in laminar microfluidics. Cryst. Growth Des. 18, 1710–1721.
Zhang, C., Dehoff, K., Hess, N., Oostrom, M., Wietsma, T.W., Valocchi, A.J., Fouke, B.W., Werth, C.J., 2010. Pore-scale study of transverse mixing induced $CaCO_3$ precipitation and permeability reduction in a model subsurface sedimentary system. Environ. Sci. Technol. 44, 7833–7838.
Zhang, Q., Deng, H., Dong, Y., Molins, S., Li, X., Steefel, C., 2022. Investigation of coupled processes in fractures and the bordering matrix via a micro-continuum reactive transport model. Water Resour. Res. 58, e2021WR030578.
Zheng, S., Xu, K., Zhang, D., 2025. Reactive precipitation during overlaying CO_2 dissolution into brine: the role of porous structure. Adv. Water Resour. 196, 104880.